

Working with Pandemic Data

This notebook evaluates seven imputation methods for filling the COVID-19 gap in the Brazil tourism arrivals time series.

Approach: select multiple non-overlapping pre-COVID windows, mask each one to simulate a gap, fit/apply each method using only the data *before* (and where applicable *after*) the masked window, and measure reconstruction error against the ground truth.

Methods compared:

1. Linear interpolation (time-based)
2. STL + PCHIP — STL decomposition fitted on pre-gap data; seasonal component projected forward into the gap; PCHIP interpolation applied to the trend only, then seasonal added back
3. Spline interpolation (order=3)
4. Auto ARIMA — forward SARIMAX fitted on pre-gap data only
5. Auto ARIMA (backward) — SARIMAX fitted on *reversed* post-gap data, predictions re-reversed to chronological order
6. ARIMA blend — linear-decay weighted average of forward and backward ARIMA forecasts
7. **AutoStat + ARIMA (fwd)** — AutoML approach: `AutoStationaryTransformer` (Joseph & Tackes, *Modern Time Series Forecasting with Python 2E*) makes the pre-gap series stationary (detrend → deseasonalize → Box-Cox with Guerrero λ , each step triggered by a statistical test), an auto-ARIMA is fitted on the stationary series, and the forecast across the gap is inverse-transformed back to arrival counts

Metrics:

- MAE, RMSE, MAPE — pointwise reconstruction error vs. ground truth
- **Variance gap** — $|\text{var}(\text{ground truth}) - \text{var}(\text{imputed})|$ over the gap, to detect methods that produce overly smooth (or overly spiky) reconstructions
- **Downstream-ARIMA RMSE / MAPE (leakage-free)** — for each method, fit an auto ARIMA on `[gap_start - 24m, 2019-06-30]` (24 months pre-gap real + 24-month imputed gap + all post-gap real up to 2019-06-30) and evaluate 6-month-ahead forecast accuracy on `2019-07-01 → 2019-12-31`. The two methods that *could* see the test window — backward ARIMA (fits on post-gap data) and spline (global cubic fit) — are re-imputed on a copy of the series truncated at 2019-06-30, so no method's gap values reflect the test region. An **oracle baseline** (auto-ARIMA fitted on the same span but with the *real* gap values instead of an imputation) is logged per window as a reference.

```
In [1]: import os
import warnings

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from pmdarima import auto_arima
from statsmodels.tsa.seasonal import STL

warnings.filterwarnings('ignore')

ROOT_PATH = os.path.dirname(os.getcwd())
DATA_FOLDER_PATH = os.path.join(ROOT_PATH, 'data', 'processed')

COVID_START = pd.Timestamp('2020-01-01')
COVID_END = pd.Timestamp('2021-12-31')

sns.set_palette('colorblind')
```

Helper Functions — Modern Time Series Forecasting with Python (2E)

The AutoML stationarity pipeline comes from the companion repository of:

Joseph, M. & Tackes, J. *Modern Time Series Forecasting with Python*, 2nd ed. (Packt, 2024) — [PacktPublishing/Modern-Time-Series-Forecasting-with-Python-2E](#), MIT License

`AutoStationaryTransformer` (`src/transforms/target_transformations.py`) chains, during `fit` , the tests explored in notebook 3:

1. `check_trend` (Mann-Kendall). If a trend is found → `DetrendingTransformer` .
2. `check_seasonality` at `m = 12`. If seasonal → `DeseasonalizingTransformer` .
3. `check_heteroscedasticity` (White). If heteroscedastic → `AddMTTransformer` when needed, then `BoxCoxTransformer` with Guerrero λ .

The fitted pipeline has an exact `inverse_transform` , which lets us forecast in the stationary space and map the forecasts back to arrival counts. The repository is expected as a sibling folder of this project (override with the `MTSF_REPO_PATH` environment variable).

```
In [2]: import sys

# Helpers credited to: Joseph, M. & Tackes, J. "Modern Time Series Forecasting with
# Python", 2nd ed. (Packt, 2024) – MIT License
# github.com/PacktPublishing/Modern-Time-Series-Forecasting-with-Python-2E
MTSF_REPO_PATH = os.getenv(
    'MTSF_REPO_PATH',
    os.path.join(
        os.path.dirname(ROOT_PATH), 'Modern-Time-Series-Forecasting-with-Python-2E'
    ),
```

```

)
if MTSF_REPO_PATH not in sys.path:
    sys.path.insert(0, MTSF_REPO_PATH)

from src.transforms.target_transformations import ( # noqa: E402
    AutoStationaryTransformer,
    BoxCoxTransformer,
)

SEASONAL_PERIOD = 12
FREQ = 'ME'

```

Data Import

```

In [3]: df = pd.read_parquet(os.path.join(DATA_FOLDER_PATH, 'train.parquet'))
df = df.set_index('date')

monthly_df = df[['arrival_count']].resample('ME').sum()

print(
    f'Series range : {monthly_df.index.min().date()} → {monthly_df.index.max().date()}
)
print(f'Total months : {len(monthly_df)}')
monthly_df.head()

```

Series range : 1989-01-31 → 2025-08-31

Total months : 440

Out[3]:

	arrival_count
date	
1989-01-31	291566.0
1989-02-28	197348.0
1989-03-31	149001.0
1989-04-30	84867.0
1989-05-31	69293.0

Imputation Benchmark

Test Windows

Thirteen pre-COVID windows of varying lengths (6 → 36 months) spanning 1995 → 2015. All windows are constrained so that the downstream-ARIMA test window stays inside the pre-COVID region (i.e. `gap_end + 48m ≤ 2019-12-31`).

The benchmark splits naturally in two groups:

- **W0 → W6** — older, non-overlapping windows scattered across historical regimes.

- **W7 → W12** — recent-year stress test: six windows of decreasing length (36m → 6m) all *ending at 2015-12-31* (the latest legal `gap_end`). These overlap heavily by construction; the goal is to measure how each method scales with gap length while holding the surrounding context fixed.

The **downstream-ARIMA metric** uses a fixed per-window geometry, anchored on each gap (all values inclusive, month-end stamps):

segment	months	offset from gap
DS train (pre-gap)	24	gap_start - 24m ... gap_start - 1m
DS gap (imputed)	6–36	gap_start ... gap_end (per-window)
DS train (post-gap)	36	gap_end + 1m ... gap_end + 36m
DS test (real)	12	gap_end + 37m ... gap_end + 48m

Resulting per-window spans:

Window	Gap	DS pre-train (24m)	DS post-train (36m)	DS test (12m)
W0	1995-01-31 → 1995-06-30 (6m)	1993-01-31 → 1994-12-31	1995-07-31 → 1998-06-30	1998-07-31 → 1999-06-30
W1	1998-01-31 → 1998-12-31 (12m)	1996-01-31 → 1997-12-31	1999-01-31 → 2001-12-31	2002-01-31 → 2002-12-31
W2	2001-01-31 → 2002-06-30 (18m)	1999-01-31 → 2000-12-31	2002-07-31 → 2005-06-30	2005-07-31 → 2006-06-30
W3	2004-01-31 → 2006-12-31 (36m)	2002-01-31 → 2003-12-31	2007-01-31 → 2009-12-31	2010-01-31 → 2010-12-31
W4	2008-01-31 → 2009-12-31 (24m)	2006-01-31 → 2007-12-31	2010-01-31 → 2012-12-31	2013-01-31 → 2013-12-31
W5	2010-01-31 → 2011-12-31 (24m)	2008-01-31 → 2009-12-31	2012-01-31 → 2014-12-31	2015-01-31 → 2015-12-31
W6	2012-01-31 → 2013-12-31 (24m)	2010-01-31 → 2011-12-31	2014-01-31 → 2016-12-31	2017-01-31 → 2017-12-31
W7	2013-01-31 → 2015-12-31 (36m)	2011-01-31 → 2012-12-31	2016-01-31 → 2018-12-31	2019-01-31 → 2019-12-31
W8	2013-07-31 → 2015-12-31 (30m)	2011-07-31 → 2013-06-30	2016-01-31 → 2018-12-31	2019-01-31 → 2019-12-31
W9	2014-01-31 → 2015-12-31 (24m)	2012-01-31 → 2013-12-31	2016-01-31 → 2018-12-31	2019-01-31 → 2019-12-31
W10	2014-07-31 → 2015-12-31 (18m)	2012-07-31 → 2014-06-30	2016-01-31 → 2018-12-31	2019-01-31 → 2019-12-31

Window	Gap	DS pre-train (24m)	DS post-train (36m)	DS test (12m)
W11	2015-01-31 → 2015-12-31 (12m)	2013-01-31 → 2014-12-31	2016-01-31 → 2018-12-31	2019-01-31 → 2019-12-31
W12	2015-07-31 → 2015-12-31 (6m)	2013-07-31 → 2015-06-30	2016-01-31 → 2018-12-31	2019-01-31 → 2019-12-31

Because W7–W12 share the same `gap_end`, they also share the same DS post-train and DS test windows — the only thing that varies is how much real data is replaced by the imputation. The DS-RMSE/DS-MAPE differences inside that cluster are therefore *purely* a function of gap length.

The reconstruction metrics use the *full* pre-COVID post-gap span (up to 2019-12-31) for backward ARIMA fitting; only the downstream-ARIMA metric uses the truncated per-window geometry above so that each method's gap imputation is leakage-free relative to its own test window. Windows are ordered chronologically by `gap_start`.

```
In [4]: WINDOWS = [
    # Older, well-spaced windows across historical regimes.
    ('W0', pd.Timestamp('1995-01-31'), pd.Timestamp('1995-06-30')), # 6m
    ('W1', pd.Timestamp('1998-01-31'), pd.Timestamp('1998-12-31')), # 12m
    ('W2', pd.Timestamp('2001-01-31'), pd.Timestamp('2002-06-30')), # 18m
    ('W3', pd.Timestamp('2004-01-31'), pd.Timestamp('2006-12-31')), # 36m
    ('W4', pd.Timestamp('2008-01-31'), pd.Timestamp('2009-12-31')), # 24m
    ('W5', pd.Timestamp('2010-01-31'), pd.Timestamp('2011-12-31')), # 24m
    ('W6', pd.Timestamp('2012-01-31'), pd.Timestamp('2013-12-31')), # 24m
    ('W7', pd.Timestamp('2013-01-31'), pd.Timestamp('2015-12-31')), # 36m
    ('W8', pd.Timestamp('2013-07-31'), pd.Timestamp('2015-12-31')), # 30m
    ('W9', pd.Timestamp('2014-01-31'), pd.Timestamp('2015-12-31')), # 24m
    ('W10', pd.Timestamp('2014-07-31'), pd.Timestamp('2015-12-31')), # 18m
    ('W11', pd.Timestamp('2015-01-31'), pd.Timestamp('2015-12-31')), # 12m
    ('W12', pd.Timestamp('2015-07-31'), pd.Timestamp('2015-12-31')), # 6m
]

RANDOM_STATE = 42 # for reproducibility of auto_arima search

for name, start, end in WINDOWS:
    n = monthly_df.loc[start:end].shape[0]
    print(f'{name}: {start.date()} → {end.date()} ({n} months)')
```

W0: 1995-01-31 → 1995-06-30 (6 months)
 W1: 1998-01-31 → 1998-12-31 (12 months)
 W2: 2001-01-31 → 2002-06-30 (18 months)
 W3: 2004-01-31 → 2006-12-31 (36 months)
 W4: 2008-01-31 → 2009-12-31 (24 months)
 W5: 2010-01-31 → 2011-12-31 (24 months)
 W6: 2012-01-31 → 2013-12-31 (24 months)
 W7: 2013-01-31 → 2015-12-31 (36 months)
 W8: 2013-07-31 → 2015-12-31 (30 months)
 W9: 2014-01-31 → 2015-12-31 (24 months)
 W10: 2014-07-31 → 2015-12-31 (18 months)
 W11: 2015-01-31 → 2015-12-31 (12 months)
 W12: 2015-07-31 → 2015-12-31 (6 months)

Helper Functions

```

In [5]: def mae(y_true, y_pred):
        return np.mean(np.abs(y_true - y_pred))

def rmse(y_true, y_pred):
    return np.sqrt(np.mean((y_true - y_pred) ** 2))

def mape(y_true, y_pred):
    mask = y_true != 0
    return np.mean(np.abs((y_true[mask] - y_pred[mask]) / y_true[mask])) * 100

def apply_linear(series_with_nan):
    return series_with_nan.interpolate(method='time').clip(lower=0)

def apply_stl_pchip(series_with_nan, gap_mask, pre_gap_end):
    "STL decomposition on pre-gap data; project seasonal forward; PCHIP the trend."
    pre_gap = series_with_nan[series_with_nan.index <= pre_gap_end].dropna()

    stl = STL(pre_gap, period=12, seasonal=13, robust=True)
    stl_res = stl.fit()

    # tile the last full year's seasonal pattern over the gap
    n_gap = int(gap_mask.sum())
    seasonal_pattern = stl_res.seasonal[-12:].values
    projected_seasonal = np.tile(seasonal_pattern, n_gap // 12 + 1)[:n_gap]

    # interpolate trend across the full series (gap included as NaN)
    trend_interp = series_with_nan.interpolate(method='pchip')

    result = series_with_nan.copy()
    result[gap_mask] = (trend_interp[gap_mask].values + projected_seasonal).clip(mi
    return result

def apply_spline(series_with_nan):
    return series_with_nan.interpolate(method='spline', order=3).clip(lower=0)
  
```

```

def apply_auto_arima(series_with_nan, gap_mask, pre_gap_end):
    """Fit SARIMAX on data strictly before the gap, forecast across the gap."""
    pre_gap = series_with_nan[series_with_nan.index <= pre_gap_end].dropna()
    n_gap = int(gap_mask.sum())
    model = auto_arima(
        pre_gap,
        seasonal=True,
        m=12,
        stepwise=False,
        n_jobs=-1,
        max_p=2,
        max_q=2,
        max_P=2,
        max_Q=2,
        d=1,
        D=0,
        information_criterion='aicc',
        out_of_sample_size=12,
        error_action='ignore',
        suppress_warnings=True,
        random_state=RANDOM_STATE,
    )
    forecast = np.asarray(model.predict(n_periods=n_gap))
    result = series_with_nan.copy()
    result[gap_mask] = np.clip(forecast, 0, None)
    return result, model, forecast

def apply_auto_arima_bwd(series_with_nan, gap_mask, post_gap_start, post_gap_end):
    """Fit SARIMAX on *reversed* post-gap data, predict backward across the gap.

    The post-gap slice is reversed to a plain numpy array (avoids index-mismatch
    during fitting); the model learns trend/seasonality moving backward in time;
    the forecast is then reversed so it aligns chronologically with the gap.
    """
    post_gap = series_with_nan[
        (series_with_nan.index >= post_gap_start)
        & (series_with_nan.index <= post_gap_end)
    ].dropna()
    n_gap = int(gap_mask.sum())

    reversed_arr = post_gap.values[::-1]
    model = auto_arima(
        reversed_arr,
        seasonal=True,
        m=12,
        stepwise=False,
        n_jobs=-1,
        max_p=2,
        max_q=2,
        max_P=2,
        max_Q=2,
        d=1,
        D=0,

```

```

        information_criterion='aicc',
        out_of_sample_size=12,
        error_action='ignore',
        suppress_warnings=True,
        random_state=RANDOM_STATE,
    )
    forecast_bwd_raw = np.asarray(model.predict(n_periods=n_gap))
    forecast_bwd = forecast_bwd_raw[::-1] # restore chronological order

    result = series_with_nan.copy()
    result[gap_mask] = np.clip(forecast_bwd, 0, None)
    return result, model, forecast_bwd

def apply_arma_blend(series_with_nan, gap_mask, forecast_fwd, forecast_bwd):
    """Linear-decay weighted average: 100% forward at gap start → 100% backward at
    n_gap = int(gap_mask.sum())
    w = np.linspace(1.0, 0.0, n_gap)
    imputed = w * forecast_fwd + (1.0 - w) * forecast_bwd

    result = series_with_nan.copy()
    result[gap_mask] = np.clip(imputed, 0, None)
    return result

# ---- Downstream-ARIMA metric (per-window Leakage-free geometry) ----
#
# All three deltas are anchored on each window's gap:
# train = [gap_start - N_DS_PRE m, gap_start - 1m] + imputed gap + [gap_end +
# test = [gap_end + N_DS_POST + 1m, gap_end + N_DS_POST + N_DS_TEST m]
#
# N_DS_POST is set to 36 (3 seasonal cycles) so the backward ARIMA's OCSB
# seasonal-differencing test has enough samples (24 is on the edge and triggers
# "no samples after first-order seasonal differencing").
N_DS_PRE = 24
N_DS_POST = 36
N_DS_TEST = 12

def ds_train_end_for(gap_end, n_post=N_DS_POST):
    "Last month included in the downstream train for a given gap."
    return gap_end + pd.DateOffset(months=n_post)

def _fit_predict_auto_arma(train_values, n_periods):
    model = auto_arma(
        train_values,
        seasonal=True,
        m=12,
        stepwise=False,
        n_jobs=-1,
        max_p=2,
        max_q=2,
        max_P=2,
        max_Q=2,
        d=1,

```

```

        D=0,
        information_criterion='aicc',
        out_of_sample_size=12,
        error_action='ignore',
        suppress_warnings=True,
        random_state=RANDOM_STATE,
    )
    return np.asarray(model.predict(n_periods=n_periods))

def downstream_arma_scores(
    full_series,
    imputed_series,
    gap_start,
    gap_end,
    n_pre=N_DS_PRE,
    n_post=N_DS_POST,
    n_test=N_DS_TEST,
    data_cutoff=None,
):
    """Train auto_arma on [gap_start - n_pre, gap_end + n_post] using real pre/post
    values plus the supplied (leakage-free) imputation for the gap itself. Forecast
    months and score against the real series.

    If `data_cutoff` is given and `test_end > data_cutoff`, the metric is skipped
    (returns NaN) – used to refuse test windows that extend into the COVID anomaly.
    """

    train_start = gap_start - pd.DateOffset(months=n_pre)
    train_end = gap_end + pd.DateOffset(months=n_post)
    test_end = train_end + pd.DateOffset(months=n_test)

    if data_cutoff is not None and test_end > data_cutoff:
        return np.nan, np.nan

    train = imputed_series.loc[train_start:train_end].copy()
    pre_idx = train.index[train.index < gap_start]
    post_idx = train.index[train.index > gap_end]
    train.loc[pre_idx] = full_series.loc[pre_idx]
    train.loc[post_idx] = full_series.loc[post_idx]

    test = full_series.loc[
        (full_series.index > train_end) & (full_series.index <= test_end)
    ]
    if len(test) < n_test or train.isna().any():
        return np.nan, np.nan

    forecast = _fit_predict_auto_arma(train.values, n_test)
    return rmse(test.values, forecast), mape(test.values, forecast)

def oracle_arma_scores(
    full_series,
    gap_start,
    gap_end,
    n_pre=N_DS_PRE,
    n_post=N_DS_POST,

```

```

n_test=N_DS_TEST,
data_cutoff=None,
):
    "Same train/test geometry as downstream_arima_scores but with the *real* gap va
    train_start = gap_start - pd.DateOffset(months=n_pre)
    train_end = gap_end + pd.DateOffset(months=n_post)
    test_end = train_end + pd.DateOffset(months=n_test)

    if data_cutoff is not None and test_end > data_cutoff:
        return np.nan, np.nan

    train = full_series.loc[train_start:train_end]
    test = full_series.loc[
        (full_series.index > train_end) & (full_series.index <= test_end)
    ]
    if len(test) < n_test:
        return np.nan, np.nan

    forecast = _fit_predict_auto_arima(train.values, n_test)
    return rmse(test.values, forecast), mape(test.values, forecast)

METHOD_STYLES = {
    'Linear': ('steelblue', '-'),
    'STL + PCHIP': ('darkorange', '-'),
    'Spline': ('seagreen', '-'),
    'Auto ARIMA (fwd)': ('mediumpurple', '-'),
    'Auto ARIMA (bwd)': ('crimson', '-'),
    'ARIMA blend': ('teal', '-'),
}

```

AutoML Helper Functions — AutoStationaryTransformer + auto-ARIMA

The seventh method applies the notebook 3 findings to imputation:

- **Fit the transformer on the pre-gap data only.** The EDA showed that λ and the trend/seasonality tests must be estimated on a stable regime. Fitting only on the pre-gap data also keeps the method leakage-free for the downstream metric, like the forward ARIMA.
- **Let auto-ARIMA choose `d` on the stationary series** (`d=None`, `max_d=1`, `D=0`). The EDA found the trend to be **stochastic** (ADF `c` vs `ct`). The transformer only removes a *deterministic* linear trend, so a unit root may remain and must still be differenced away. The forward/backward ARIMA methods keep their fixed `d=1`.
- **Map the forecast back with `inverse_transform`.** The detrender extends the linear trend and the deseasonalizer repeats the period averages over the forecast dates, then the Box-Cox and offset steps are undone.

In [6]: *# Helper `AutoStationaryTransformer` from Joseph & Tackes, "Modern Time Series Forecasting with Python" 2E – src/transforms/target_transformations.py*

```
def fit_auto_stationary(train_series):
```

```

"""Fit a fresh AutoStationaryTransformer (Mann-Kendall trend, Guerrero Box-Cox)
# fresh dicts every fit: the transformer mutates its parameter dicts in place
ast = AutoStationaryTransformer(
    confidence=0.05,
    seasonal_period=SEASONAL_PERIOD,
    trend_check_params={'mann_kendall': True},
    detrender_params={'degree': 1},
    deseasonalizer_params={},
    box_cox_params={'optimization': 'guerrero'},
)
return ast.fit(train_series, freq=FREQ)

def describe_pipeline(ast):
    steps = [type(tr).__name__.replace('Transformer', '') for tr in ast._pipeline]
    return ' → '.join(steps) if steps else '(none)'

def fit_predict_auto_stationary_arima(train_series, n_periods):
    """AutoStationaryTransformer → auto-ARIMA on the stationary series → inverse."""
    ast = fit_auto_stationary(train_series)
    stationary = ast.transform(train_series)
    model = auto_arima(
        stationary.to_numpy(),
        seasonal=True,
        m=12,
        stepwise=False,
        n_jobs=-1,
        max_p=2,
        max_q=2,
        max_P=2,
        max_Q=2,
        d=None,
        max_d=1,
        D=0,
        information_criterion='aicc',
        out_of_sample_size=12,
        error_action='ignore',
        suppress_warnings=True,
        random_state=RANDOM_STATE,
    )
    forecast_index = pd.date_range(
        train_series.index[-1] + pd.offsets.MonthEnd(1), periods=n_periods, freq=FREQ
    )
    forecast = ast.inverse_transform(
        pd.Series(np.asarray(model.predict(n_periods=n_periods)), index=forecast_index)
    )
    return np.clip(forecast.to_numpy(), 0, None), ast, model

def apply_auto_stationary_arima(series_with_nan, gap_mask, pre_gap_end):
    """Fit the AutoML pipeline on data strictly before the gap, forecast across it.
    pre_gap = series_with_nan[series_with_nan.index <= pre_gap_end].dropna()
    n_gap = int(gap_mask.sum())
    forecast, ast, model = fit_predict_auto_stationary_arima(pre_gap, n_gap)
    result = series_with_nan.copy()

```

```

result[gap_mask] = forecast
return result, ast, model

```

```
METHOD_STYLES['AutoStat + ARIMA (fwd)'] = ('saddlebrown', '-')

```

Run Benchmark Across All Windows

```

In [7]: # ---- Console formatting helpers -----
# A single column spec drives both the metrics-table header and its rows, so the
# two can never drift out of alignment. Each column is (header, header_align,
# cell_formatter); the cell_formatter renders one value to a fixed-width string.
def _fmt_int(width):
    return lambda v: f'{v:>{width},.0f}'

def _fmt_int_na(width):
    "Integer, right-aligned; non-finite values render as 'n/a'."
    return lambda v: f'{v:>{width},.0f}' if np.isfinite(v) else f'{"n/a":>{width}}'

def _fmt_ratio(width):
    return lambda v: f'{v:>{width}.3f}'

def _fmt_pct(width):
    "Percent with trailing %, right-aligned; non-finite values render as 'n/a'."
    return lambda v: f'{v:>{width - 1}.2f}%' if np.isfinite(v) else f'{"n/a":>{width}}'

METRIC_COLUMNS = [
    ('Method', '<17', lambda v: f'{v:<17}'),
    ('MAE', '>10', _fmt_int(10)),
    ('RMSE', '>10', _fmt_int(10)),
    ('MAPE', '>7', _fmt_pct(7)),
    ('orig_var', '>14', _fmt_int(14)),
    ('pred_var', '>14', _fmt_int(14)),
    ('var_ratio', '>10', _fmt_ratio(10)),
    ('DS-RMSE', '>10', _fmt_int_na(10)),
    ('DS-MAPE', '>8', _fmt_pct(8)),
]

def print_banner(title):
    "Print a window section header bracketed by rules."
    rule = '=' * 55
    print(f'\n{rule}\n {title}\n{rule}')

def metric_header():
    cells = (f'{label:{align}}' for label, align, _ in METRIC_COLUMNS)
    return ' ' + ' '.join(cells)

def metric_row(values):

```

```

    "Render one row; `values` aligns positionally with METRIC_COLUMNS."
    cells = (fmt(v) for (_, _, fmt), v in zip(METRIC_COLUMNS, values))
    return ' ' + ' '.join(cells)

# ---- Benchmark -----
# ---- Methodology Legend (printed once before the per-window run) -----
# Explains what each window block below reports, so the log is self-documenting.
print(f'{"=" * 75}')
print('  BENCHMARK METHODOLOGY')
print(f'{"=" * 75}')
print(
    """
    For each pre-COVID window we mask the gap, impute it with seven methods, and
    score the result two ways:

    RECONSTRUCTION metrics - how close the imputation is to the real gap values.
    MAE / RMSE / MAPE    pointwise error vs. ground truth over the gap.
    orig_var / pred_var  variance of the true gap vs. the imputed gap.
    var_ratio            = pred_var / orig_var.  1.0 = ideal; <1 over-smooth
                        (flattened), >1 over-spiky.

    DOWNSTREAM (DS) metrics - does the imputation help a model forecast the FUTURE?
    Fit an auto-ARIMA on [24m real pre-gap] + [imputed gap] + [36m real post-gap],
    then forecast 12 months and score against the real series.
    DS-RMSE / DS-MAPE    error of that 12-month forecast.
    Built leakage-free: methods that could otherwise peek at the test window
    (backward ARIMA, spline) are re-imputed on a series truncated at the DS
    train end, so no gap value reflects the test region.
    'n/a' = the window's DS test would fall inside COVID, so the metric is skipped.

    ORACLE baseline - same DS train/test geometry but using the REAL gap values
    instead of an imputation. It is the best DS score any imputation could reach;
    treat it as the target each method is chasing, not a competitor.
    """
)

records = []
window_results = {} # stores imputed series per window for plotting

# Cap post-gap training data at the pre-COVID boundary so the backward model
# never sees the pandemic anomaly (used only by the reconstruction metric).
POST_GAP_CUTOFF = COVID_START - pd.Timedelta(days=1)

for win_name, gap_start, gap_end in WINDOWS:
    print_banner(f'{win_name}: {gap_start.date()} → {gap_end.date()}')

    gap_mask = (monthly_df.index >= gap_start) & (monthly_df.index <= gap_end)
    truth = monthly_df.loc[gap_mask, 'arrival_count'].values
    truth_var = float(np.var(truth))

    masked = monthly_df['arrival_count'].copy()
    masked[gap_mask] = np.nan

    # Training cutoffs
    pre_gap_start = monthly_df.index[0]

```

```

pre_gap_end = monthly_df[monthly_df.index < gap_start].index[-1]
post_gap_start = monthly_df[monthly_df.index > gap_end].index[0]
post_gap_end = POST_GAP_CUTOFF
n_pre = monthly_df.loc[pre_gap_start:pre_gap_end].shape[0]
n_post = monthly_df.loc[post_gap_start:post_gap_end].shape[0]

# Downstream-metric geometry (per-window, anchored on gap_end)
ds_train_end = ds_train_end_for(gap_end)
ds_test_end = ds_train_end + pd.DateOffset(months=N_DS_TEST)
ds_in_pre_covid = ds_test_end <= POST_GAP_CUTOFF
ds_test_start = ds_train_end + pd.DateOffset(days=1)
ds_status = '' if ds_in_pre_covid else 'test would enter COVID - metric N/A'

print(
    f' Pre-gap training span: {pre_gap_start.date()} → {pre_gap_end.date()} (
)
print(
    f' Post-gap training span: {post_gap_start.date()} → {post_gap_end.date()}
)
print(
    f' Downstream geometry: train ends {ds_train_end.date()} | '
    f'test {ds_test_start.date()} → {ds_test_end.date()} '
    f'({N_DS_PRE}m pre + {N_DS_POST}m post + {N_DS_TEST}m test){ds_status}'
)
print(f' Ground-truth variance: {truth_var:,.0f}')

# ---- Imputations used for the reconstruction metrics (full post-gap context)
s_linear = apply_linear(masked)
pred_linear = s_linear[gap_mask].values

s_stl_pchip = apply_stl_pchip(masked, gap_mask, pre_gap_end)
pred_stl_pchip = s_stl_pchip[gap_mask].values

s_spline = apply_spline(masked)
pred_spline = s_spline[gap_mask].values

s_arima_fwd, arima_fwd_model, forecast_fwd = apply_auto_arima(
    masked, gap_mask, pre_gap_end
)
pred_arima_fwd = s_arima_fwd[gap_mask].values
print(
    f' ARIMA fwd order: {arima_fwd_model.order} x {arima_fwd_model.seas
)

s_arima_bwd, arima_bwd_model, forecast_bwd = apply_auto_arima_bwd(
    masked, gap_mask, post_gap_start, post_gap_end
)
pred_arima_bwd = s_arima_bwd[gap_mask].values
print(
    f' ARIMA bwd order: {arima_bwd_model.order} x {arima_bwd_model.seas
)

s_arima_blend = apply_arima_blend(masked, gap_mask, forecast_fwd, forecast_bwd)
pred_arima_blend = s_arima_blend[gap_mask].values

# AutoML: AutoStationaryTransformer fitted on pre-gap data only (Leakage-free)

```

```

s_ast, ast_fitted, ast_arima_model = apply_auto_stationary_arima(
    masked, gap_mask, pre_gap_end
)
pred_ast = s_ast[gap_mask].values
print(
    f' AutoStat pipeline:      {describe_pipeline(ast_fitted)} | '
    f'ARIMA {ast_arima_model.order} x {ast_arima_model.seasonal_order}'
)

# ---- Leakage-free imputations for the downstream metric ----
# Only build these if the test window fits inside the pre-COVID region; otherwise
# there's no downstream metric to compute for this window.
if ds_in_pre_covid:
    masked_lf = masked.copy()
    masked_lf.loc[masked_lf.index > ds_train_end] = np.nan

    s_spline_lf = apply_spline(masked_lf)
    _, _, forecast_bwd_lf = apply_auto_arima_bwd(
        masked_lf, gap_mask, post_gap_start, ds_train_end
    )
    s_arima_bwd_lf = masked.copy()
    s_arima_bwd_lf[gap_mask] = np.clip(forecast_bwd_lf, 0, None)
    s_arima_blend_lf = apply_arima_blend(
        masked, gap_mask, forecast_fwd, forecast_bwd_lf
    )
else:
    s_spline_lf = s_arima_bwd_lf = s_arima_blend_lf = None

# Store imputed series for plotting (reconstruction-side imputations)
window_results[win_name] = {
    'gap_mask': gap_mask,
    'Linear': s_linear,
    'STL + PCHIP': s_stl_pchip,
    'Spline': s_spline,
    'Auto ARIMA (fwd)': s_arima_fwd,
    'Auto ARIMA (bwd)': s_arima_bwd,
    'ARIMA blend': s_arima_blend,
    'AutoStat + ARIMA (fwd)': s_ast,
}

# ---- Oracle baseline (real gap values, same downstream geometry) ----
oracle_rmse, oracle_mape = oracle_arima_scores(
    monthly_df['arrival_count'], gap_start, gap_end, data_cutoff=POST_GAP_CUTOFF
)
if np.isfinite(oracle_rmse):
    print(
        f' Oracle baseline (real gap)          '
        f'DS-RMSE={_fmt_int_na(10)(oracle_rmse)} DS-MAPE={_fmt_pct(7)(oracle_m'
    )
else:
    print(' Oracle baseline (real gap)          N/A')

# ---- Per-method metrics ----
# (method_name, gap-reconstruction prediction, imputed series for the Leakage-f
# downstream metric – may be None when the window's DS test extends into COVID)
method_preds = [

```

```

('Linear', pred_linear, s_linear),
('STL + PCHIP', pred_stl_pchip, s_stl_pchip),
('Spline', pred_spline, s_spline_1f),
('Auto ARIMA (fwd)', pred_arima_fwd, s_arima_fwd),
('Auto ARIMA (bwd)', pred_arima_bwd, s_arima_bwd_1f),
('ARIMA blend', pred_arima_blend, s_arima_blend_1f),
('AutoStat + ARIMA (fwd)', pred_ast, s_ast),
]

print('\n' + metric_header())
for method_name, pred, s_imputed_1f in method_preds:
    mae_v = mae(truth, pred)
    rmse_v = rmse(truth, pred)
    mape_v = mape(truth, pred)
    pred_var = float(np.var(pred))
    var_ratio = pred_var / truth_var if truth_var > 0 else np.nan

    if s_imputed_1f is not None:
        ds_rmse, ds_mape = downstream_arima_scores(
            monthly_df['arrival_count'],
            s_imputed_1f,
            gap_start,
            gap_end,
            data_cutoff=POST_GAP_CUTOFF,
        )
    else:
        ds_rmse, ds_mape = np.nan, np.nan

    records.append({
        'Window': win_name,
        'Method': method_name,
        'MAE': mae_v,
        'RMSE': rmse_v,
        'MAPE': mape_v,
        'OrigVar': truth_var,
        'PredVar': pred_var,
        'VarRatio': var_ratio,
        'DownstreamRMSE': ds_rmse,
        'DownstreamMAPE': ds_mape,
        'OracleRMSE': oracle_rmse,
        'OracleMAPE': oracle_mape,
    })
print(
    metric_row([
        method_name,
        mae_v,
        rmse_v,
        mape_v,
        truth_var,
        pred_var,
        var_ratio,
        ds_rmse,
        ds_mape,
    ])
)

```

```
results_df = pd.DataFrame(records)
```

```
=====
BENCHMARK METHODOLOGY
=====
```

For each pre-COVID window we mask the gap, impute it with seven methods, and score the result two ways:

RECONSTRUCTION metrics - how close the imputation is to the real gap values.

MAE / RMSE / MAPE pointwise error vs. ground truth over the gap.
orig_var / pred_var variance of the true gap vs. the imputed gap.
var_ratio = pred_var / orig_var. 1.0 = ideal; <1 over-smooth
(flattened), >1 over-spiky.

DOWNSTREAM (DS) metrics - does the imputation help a model forecast the FUTURE?

Fit an auto-ARIMA on [24m real pre-gap] + [imputed gap] + [36m real post-gap], then forecast 12 months and score against the real series.

DS-RMSE / DS-MAPE error of that 12-month forecast.

Built leakage-free: methods that could otherwise peek at the test window (backward ARIMA, spline) are re-imputed on a series truncated at the DS train end, so no gap value reflects the test region.

'n/a' = the window's DS test would fall inside COVID, so the metric is skipped.

ORACLE baseline - same DS train/test geometry but using the REAL gap values instead of an imputation. It is the best DS score any imputation could reach; treat it as the target each method is chasing, not a competitor.

```
=====
W0: 1995-01-31 → 1995-06-30
=====
```

Pre-gap training span: 1989-01-31 → 1994-12-31 (72 months)

Post-gap training span: 1995-07-31 → 2019-12-31 (294 months)

Downstream geometry: train ends 1998-06-30 | test 1998-07-01 → 1999-06-30 (2
4m pre + 36m post + 12m test)

Ground-truth variance: 5,749,626,530

ARIMA fwd order: (0, 1, 0) x (1, 0, 1, 12)
 ARIMA bwd order: (2, 1, 2) x (1, 0, 0, 12)
 AutoStat pipeline: Detrending → Deseasonalizing | ARIMA (1, 0, 0) x (2, 0, 2, 12)

Oracle baseline (real gap) DS-RMSE= 39,783 DS-MAPE= 8.75%

Method	MAE	RMSE	MAPE	orig_var	pred_var
var_ratio DS-RMSE DS-MAPE					
Linear	57,966	79,452	27.59%	5,749,626,530	32,975,435
0.006 54,505 11.11%					
STL + PCHIP	35,091	40,473	21.40%	5,749,626,530	13,264,083,930
2.307 39,746 8.77%					
Spline	44,930	64,370	22.35%	5,749,626,530	335,770,679
0.058 77,306 10.21%					
Auto ARIMA (fwd)	10,675	12,593	5.67%	5,749,626,530	4,429,791,512
0.770 39,756 8.76%					
Auto ARIMA (bwd)	24,246	27,818	13.62%	5,749,626,530	3,998,315,379
0.695 39,774 8.87%					
ARIMA blend	14,527	15,826	9.09%	5,749,626,530	4,819,220,718
0.838 39,816 8.74%					
AutoStat + ARIMA (fwd)	31,969	33,406	19.02%	5,749,626,530	5,129,12
2,229 0.892 39,766 8.76%					

=====

W1: 1998-01-31 → 1998-12-31

=====

Pre-gap training span: 1989-01-31 → 1997-12-31 (108 months)
 Post-gap training span: 1999-01-31 → 2019-12-31 (252 months)
 Downstream geometry: train ends 2001-12-31 | test 2002-01-01 → 2002-12-31 (2
 4m pre + 36m post + 12m test)
 Ground-truth variance: 16,739,711,911
 ARIMA fwd order: (0, 1, 0) x (1, 0, 1, 12)
 ARIMA bwd order: (2, 1, 2) x (1, 0, 0, 12)
 AutoStat pipeline: Detrending → Deseasonalizing → AddM → BoxCox | ARIMA (1,
 0, 0) x (1, 0, 2, 12)

Oracle baseline (real gap) DS-RMSE= 53,576 DS-MAPE= 13.11%

Method	MAE	RMSE	MAPE	orig_var	pred_var
var_ratio DS-RMSE DS-MAPE					
Linear	230,815	245,635	59.77%	16,739,711,911	15,674,176,745
0.936 49,978 13.09%					
STL + PCHIP	208,101	227,161	56.86%	16,739,711,911	12,699,961,388
0.759 48,513 11.26%					
Spline	382,538	407,890	108.06%	16,739,711,911	40,192,921,050
2.401 71,037 22.41%					
Auto ARIMA (fwd)	148,944	164,447	35.63%	16,739,711,911	4,006,069,104
0.239 51,766 13.56%					
Auto ARIMA (bwd)	80,697	88,398	24.21%	16,739,711,911	7,757,415,195
0.463 59,947 13.43%					
ARIMA blend	89,540	126,147	18.60%	16,739,711,911	4,057,694,100
0.242 50,660 13.30%					
AutoStat + ARIMA (fwd)	182,561	196,931	44.64%	16,739,711,911	4,150,78
9,367 0.248 69,012 17.75%					

=====

W2: 2001-01-31 → 2002-06-30

```

=====
Pre-gap training span: 1989-01-31 → 2000-12-31 (144 months)
Post-gap training span: 2002-07-31 → 2019-12-31 (210 months)
Downstream geometry: train ends 2005-06-30 | test 2005-07-01 → 2006-06-30 (2
4m pre + 36m post + 12m test)
Ground-truth variance: 18,789,422,600
ARIMA fwd order: (2, 1, 1) x (1, 0, 1, 12)
ARIMA bwd order: (2, 1, 2) x (1, 0, 0, 12)
AutoStat pipeline: Detrending → Deseasonalizing → AddM → BoxCox | ARIMA (2,
0, 0) x (1, 0, 0, 12)
Oracle baseline (real gap) DS-RMSE= 58,258 DS-MAPE= 8.97%

```

Method	MAE	RMSE	MAPE	orig_var	pred_var
var_ratio DS-RMSE DS-MAPE					
Linear	97,408	115,953	26.93%	18,789,422,600	1,890,053,128
0.101 71,965 10.82%					
STL + PCHIP	66,504	90,296	17.55%	18,789,422,600	32,110,631,227
1.709 67,266 10.97%					
Spline	170,270	196,108	53.20%	18,789,422,600	9,683,684,098
0.515 65,827 10.31%					
Auto ARIMA (fwd)	147,751	174,212	47.72%	18,789,422,600	15,403,196,199
0.820 59,637 10.01%					
Auto ARIMA (bwd)	104,061	157,821	22.82%	18,789,422,600	2,531,188,845
0.135 100,908 12.96%					
ARIMA blend	41,469	46,479	13.55%	18,789,422,600	15,127,747,452
0.805 71,432 11.01%					
AutoStat + ARIMA (fwd)	87,223	103,226	28.20%	18,789,422,600	11,284,016,309
0.601 75,963 12.33%					

```

=====
W3: 2004-01-31 → 2006-12-31
=====
Pre-gap training span: 1989-01-31 → 2003-12-31 (180 months)
Post-gap training span: 2007-01-31 → 2019-12-31 (156 months)
Downstream geometry: train ends 2009-12-31 | test 2010-01-01 → 2010-12-31 (2
4m pre + 36m post + 12m test)
Ground-truth variance: 15,076,159,859
ARIMA fwd order: (2, 1, 1) x (2, 0, 0, 12)
ARIMA bwd order: (2, 1, 2) x (1, 0, 0, 12)
AutoStat pipeline: Detrending → Deseasonalizing → AddM → BoxCox | ARIMA (1,
0, 0) x (2, 0, 2, 12)
Oracle baseline (real gap) DS-RMSE= 113,316 DS-MAPE= 24.39%

```

Method	MAE	RMSE	MAPE	orig_var	pred_var
var_ratio DS-RMSE DS-MAPE					
Linear	206,764	235,632	55.25%	15,076,159,859	10,868,102,175
0.721 148,627 30.50%					
STL + PCHIP	242,680	262,413	62.22%	15,076,159,859	21,233,943,987
1.408 134,754 29.76%					
Spline	1,634,734	1,756,833	417.77%	15,076,159,859	420,568,922,188
8 27.896 120,006 16.31%					
Auto ARIMA (fwd)	61,721	83,953	15.53%	15,076,159,859	7,436,009,999
0.493 83,783 14.73%					
Auto ARIMA (bwd)	223,473	253,744	60.37%	15,076,159,859	12,390,970,613
0.822 77,927 14.38%					
ARIMA blend	96,237	105,908	25.53%	15,076,159,859	5,817,022,473

0.386	88,877	14.42%				
AutoStat + ARIMA (fwd)	53,500	68,361	13.86%	15,076,159,859	11,457,93	
1,889	0.760	77,323	13.44%			

=====

W4: 2008-01-31 → 2009-12-31

=====

Pre-gap training span: 1989-01-31 → 2007-12-31 (228 months)
 Post-gap training span: 2010-01-31 → 2019-12-31 (120 months)
 Downstream geometry: train ends 2012-12-31 | test 2013-01-01 → 2013-12-31 (2
 4m pre + 36m post + 12m test)
 Ground-truth variance: 18,227,732,347
 ARIMA fwd order: (2, 1, 2) x (1, 0, 0, 12)
 ARIMA bwd order: (1, 1, 2) x (1, 0, 0, 12)
 AutoStat pipeline: Detrending → Deseasonalizing → AddM → BoxCox | ARIMA (2,
 0, 0) x (1, 0, 0, 12)
 Oracle baseline (real gap) DS-RMSE= 94,891 DS-MAPE= 20.76%

Method	MAE	RMSE	MAPE	orig_var	pred_var
var_ratio DS-RMSE DS-MAPE					
Linear	222,565	241,555	63.82%	18,227,732,347	2,971,053,353
0.163 95,504 19.88%					
STL + PCHIP	222,444	235,511	61.48%	18,227,732,347	15,177,535,416
0.833 76,769 16.37%					
Spline	590,030	624,727	166.69%	18,227,732,347	25,686,516,525
1.409 125,022 27.16%					
Auto ARIMA (fwd)	70,499	84,634	20.15%	18,227,732,347	9,075,278,097
0.498 88,315 19.46%					
Auto ARIMA (bwd)	162,187	175,666	46.60%	18,227,732,347	3,977,622,270
0.218 177,460 39.38%					
ARIMA blend	103,658	113,436	29.71%	18,227,732,347	6,214,831,203
0.341 92,310 20.11%					
AutoStat + ARIMA (fwd)	67,827	80,298	19.97%	18,227,732,347	11,752,76
4,479 0.645 88,147 19.47%					

=====

W5: 2010-01-31 → 2011-12-31

=====

Pre-gap training span: 1989-01-31 → 2009-12-31 (252 months)
 Post-gap training span: 2012-01-31 → 2019-12-31 (96 months)
 Downstream geometry: train ends 2014-12-31 | test 2015-01-01 → 2015-12-31 (2
 4m pre + 36m post + 12m test)
 Ground-truth variance: 13,845,560,200
 ARIMA fwd order: (2, 1, 2) x (1, 0, 0, 12)
 ARIMA bwd order: (2, 1, 1) x (1, 0, 0, 12)
 AutoStat pipeline: Detrending → Deseasonalizing → AddM → BoxCox | ARIMA (1,
 1, 1) x (1, 0, 0, 12)
 Oracle baseline (real gap) DS-RMSE= 219,691 DS-MAPE= 43.92%

Method	MAE	RMSE	MAPE	orig_var	pred_var
var_ratio DS-RMSE DS-MAPE					
Linear	185,566	206,143	46.43%	13,845,560,200	10,113,830,792
0.730 225,048 47.30%					
STL + PCHIP	141,282	169,954	32.98%	13,845,560,200	26,006,283,811
1.878 239,745 52.58%					
Spline	695,990	777,615	172.14%	13,845,560,200	155,177,542,72

8	11.208	356,058	81.01%				
Auto ARIMA (fwd)		49,988	63,397	10.48%	13,845,560,200	8,749,044,183	
0.632	223,324	45.23%					
Auto ARIMA (bwd)		213,637	238,893	56.71%	13,845,560,200	5,921,534,197	
0.428	311,910	70.38%					
ARIMA blend		86,318	97,479	22.15%	13,845,560,200	6,648,837,577	
0.480	250,159	53.27%					
AutoStat + ARIMA (fwd)		41,124	54,712	8.65%	13,845,560,200	10,413,97	
8,970	0.752	224,337	45.21%				

=====

W6: 2012-01-31 → 2013-12-31

=====

Pre-gap training span: 1989-01-31 → 2011-12-31 (276 months)
Post-gap training span: 2014-01-31 → 2019-12-31 (72 months)
Downstream geometry: train ends 2016-12-31 | test 2017-01-01 → 2017-12-31 (2
4m pre + 36m post + 12m test)
Ground-truth variance: 15,687,774,814
ARIMA fwd order: (2, 1, 2) x (1, 0, 0, 12)
ARIMA bwd order: (2, 1, 1) x (0, 0, 1, 12)
AutoStat pipeline: Detrending → Deseasonalizing → AddM → BoxCox | ARIMA (1,
1, 1) x (0, 0, 2, 12)
Oracle baseline (real gap) DS-RMSE= 191,332 DS-MAPE= 34.35%

Method	MAE	RMSE	MAPE	orig_var	pred_var
var_ratio	DS-RMSE	DS-MAPE			
Linear	133,815	156,180	32.99%	15,687,774,814	29,454,818
0.002	197,435	35.36%			
STL + PCHIP	100,481	106,981	23.17%	15,687,774,814	13,898,677,192
0.886	200,475	36.52%			
Spline	246,795	287,357	61.05%	15,687,774,814	8,606,701,938
0.549	207,450	36.70%			
Auto ARIMA (fwd)	81,736	96,631	19.86%	15,687,774,814	7,368,965,086
0.470	198,141	35.97%			
Auto ARIMA (bwd)	221,439	259,160	54.42%	15,687,774,814	10,849,883,830
0.692	173,240	28.27%			
ARIMA blend	127,213	155,031	31.63%	15,687,774,814	5,346,862,819
0.341	201,836	35.79%			
AutoStat + ARIMA (fwd)	43,066	54,572	10.34%	15,687,774,814	11,078,94
2,224	0.706	199,056	36.35%		

=====

W7: 2013-01-31 → 2015-12-31

=====

Pre-gap training span: 1989-01-31 → 2012-12-31 (288 months)
Post-gap training span: 2016-01-31 → 2019-12-31 (48 months)
Downstream geometry: train ends 2018-12-31 | test 2019-01-01 → 2019-12-31 (2
4m pre + 36m post + 12m test)
Ground-truth variance: 33,387,679,044
ARIMA fwd order: (2, 1, 2) x (1, 0, 0, 12)
ARIMA bwd order: (0, 1, 0) x (0, 0, 2, 12)
AutoStat pipeline: Detrending → Deseasonalizing → AddM → BoxCox | ARIMA (1,
1, 1) x (0, 0, 2, 12)
Oracle baseline (real gap) DS-RMSE= 117,749 DS-MAPE= 24.05%

Method	MAE	RMSE	MAPE	orig_var	pred_var
--------	-----	------	------	----------	----------

var_ratio	DS-RMSE	DS-MAPE					
Linear		339,673	390,284	83.58%	33,387,679,044	19,696,959,494	
0.590	106,920	21.71%					
STL + PCHIP		431,438	461,795	99.42%	33,387,679,044	28,826,142,984	
0.863	50,320	9.04%					
Spline		1,756,150	1,898,137	392.10%	33,387,679,044	507,334,716,25	
6	15.195	299,527	61.70%				
Auto ARIMA (fwd)		132,798	168,708	30.13%	33,387,679,044	9,075,656,434	
0.272	61,011	11.24%					
Auto ARIMA (bwd)		668,386	726,929	155.08%	33,387,679,044	53,270,405,698	
1.596	130,305	26.72%					
ARIMA blend		297,458	334,959	71.85%	33,387,679,044	16,026,182,499	
0.480	67,519	12.69%					
AutoStat + ARIMA (fwd)		104,838	152,939	21.38%	33,387,679,044	10,888,90	
2,421	0.326	82,873	16.04%				

=====

W8: 2013-07-31 → 2015-12-31

=====

Pre-gap training span: 1989-01-31 → 2013-06-30 (294 months)
Post-gap training span: 2016-01-31 → 2019-12-31 (48 months)
Downstream geometry: train ends 2018-12-31 | test 2019-01-01 → 2019-12-31 (2
4m pre + 36m post + 12m test)
Ground-truth variance: 35,218,193,321
ARIMA fwd order: (2, 1, 2) x (1, 0, 0, 12)
ARIMA bwd order: (0, 1, 0) x (0, 0, 2, 12)
AutoStat pipeline: Detrending → Deseasonalizing → AddM → BoxCox | ARIMA (1,
1, 1) x (1, 0, 0, 12)
Oracle baseline (real gap) DS-RMSE= 55,954 DS-MAPE= 8.13%

Method	MAE	RMSE	MAPE	orig_var	pred_var
var_ratio	DS-RMSE	DS-MAPE			
Linear		266,253	333,361	64.39%	35,218,193,321
1.188	96,886	19.53%			41,850,738,996
STL + PCHIP		288,906	347,137	65.23%	35,218,193,321
2.029	80,202	15.32%			71,452,382,586
Spline		1,054,540	1,241,535	238.68%	35,218,193,321
1	12.726	260,662	53.25%		448,186,025,85
Auto ARIMA (fwd)		119,258	171,764	24.49%	35,218,193,321
0.236	75,707	14.04%			8,315,386,128
Auto ARIMA (bwd)		608,112	665,084	142.89%	35,218,193,321
1.212	151,452	30.05%			42,687,576,109
ARIMA blend		268,626	303,719	64.74%	35,218,193,321
0.541	80,237	15.44%			19,060,796,994
AutoStat + ARIMA (fwd)		108,828	169,322	20.37%	35,218,193,321
0,821	0.277	74,743	13.71%		9,753,06

=====

W9: 2014-01-31 → 2015-12-31

=====

Pre-gap training span: 1989-01-31 → 2013-12-31 (300 months)
Post-gap training span: 2016-01-31 → 2019-12-31 (48 months)
Downstream geometry: train ends 2018-12-31 | test 2019-01-01 → 2019-12-31 (2
4m pre + 36m post + 12m test)
Ground-truth variance: 40,691,166,990
ARIMA fwd order: (2, 1, 2) x (1, 0, 0, 12)

ARIMA bwd order: (0, 1, 0) x (0, 0, 2, 12)
 AutoStat pipeline: Detrending → Deseasonalizing → AddM → BoxCox | ARIMA (1, 1, 1) x (0, 0, 2, 12)
 Oracle baseline (real gap) DS-RMSE= 127,319 DS-MAPE= 26.13%

Method		MAE	RMSE	MAPE	orig_var	pred_var
var_ratio	DS-RMSE	DS-MAPE				
Linear		338,143	393,287	83.15%	40,691,166,990	19,014,024,915
0.467	66,418	8.79%				
STL + PCHIP		425,936	450,904	97.68%	40,691,166,990	26,642,340,484
0.655	48,968	8.88%				
Spline		1,293,633	1,402,721	292.41%	40,691,166,990	251,799,449,505
5	6.188	227,574	47.46%			
Auto ARIMA (fwd)		129,440	180,646	25.16%	40,691,166,990	10,012,082,125
0.246	59,275	11.15%				
Auto ARIMA (bwd)		524,119	572,136	122.86%	40,691,166,990	29,077,870,168
0.715	92,076	18.84%				
ARIMA blend		251,861	289,896	60.67%	40,691,166,990	12,642,674,408
0.311	51,179	9.29%				
AutoStat + ARIMA (fwd)		122,834	180,980	22.96%	40,691,166,990	12,421,477,326
0.305	73,294	13.87%				

=====

W10: 2014-07-31 → 2015-12-31

=====

Pre-gap training span: 1989-01-31 → 2014-06-30 (306 months)
 Post-gap training span: 2016-01-31 → 2019-12-31 (48 months)
 Downstream geometry: train ends 2018-12-31 | test 2019-01-01 → 2019-12-31 (24m pre + 36m post + 12m test)
 Ground-truth variance: 35,713,167,317
 ARIMA fwd order: (2, 1, 2) x (1, 0, 0, 12)
 ARIMA bwd order: (0, 1, 0) x (0, 0, 2, 12)
 AutoStat pipeline: Detrending → Deseasonalizing | ARIMA (2, 1, 2) x (1, 0, 0, 12)
 Oracle baseline (real gap) DS-RMSE= 62,349 DS-MAPE= 8.17%

Method		MAE	RMSE	MAPE	orig_var	pred_var
var_ratio	DS-RMSE	DS-MAPE				
Linear		522,061	556,241	124.74%	35,713,167,317	317,438,579
0.009	144,991	30.00%				
STL + PCHIP		528,922	541,890	120.02%	35,713,167,317	12,369,074,219
0.346	121,995	24.97%				
Spline		2,486,161	2,617,000	540.45%	35,713,167,317	650,718,087,885
5	18.221	510,409	105.46%			
Auto ARIMA (fwd)		215,372	291,214	50.79%	35,713,167,317	37,688,239,333
1.055	81,167	16.17%				
Auto ARIMA (bwd)		462,929	498,381	107.52%	35,713,167,317	21,673,313,024
0.607	124,628	24.69%				
ARIMA blend		262,509	314,056	66.04%	35,713,167,317	12,103,175,068
0.339	100,520	20.32%				
AutoStat + ARIMA (fwd)		172,778	222,147	41.68%	35,713,167,317	18,523,142,146
0.519	78,061	15.47%				

=====

W11: 2015-01-31 → 2015-12-31

=====

Pre-gap training span: 1989-01-31 → 2014-12-31 (312 months)
 Post-gap training span: 2016-01-31 → 2019-12-31 (48 months)
 Downstream geometry: train ends 2018-12-31 | test 2019-01-01 → 2019-12-31 (2
 4m pre + 36m post + 12m test)
 Ground-truth variance: 37,115,588,103
 ARIMA fwd order: (2, 1, 2) x (1, 0, 0, 12)
 ARIMA bwd order: (0, 1, 0) x (0, 0, 2, 12)
 AutoStat pipeline: Detrending → Deseasonalizing | ARIMA (1, 1, 2) x (1, 0,
 0, 12)

Oracle baseline (real gap) DS-RMSE= 144,430 DS-MAPE= 29.85%

Method	MAE	RMSE	MAPE	orig_var	pred_var
var_ratio DS-RMSE DS-MAPE					
Linear	449,459	492,705	112.74%	37,115,588,103	3,820,430,417
0.103 80,784 15.56%					
STL + PCHIP	496,627	512,525	116.95%	37,115,588,103	10,228,315,296
0.276 65,298 12.05%					
Spline	1,324,689	1,412,601	322.07%	37,115,588,103	94,411,953,499
2.544 262,340 54.38%					
Auto ARIMA (fwd)	183,739	249,417	45.44%	37,115,588,103	29,334,596,704
0.790 107,124 21.21%					
Auto ARIMA (bwd)	383,998	403,884	92.61%	37,115,588,103	11,503,076,659
0.310 94,484 19.19%					
ARIMA blend	269,915	314,363	68.89%	37,115,588,103	13,092,027,646
0.353 168,037 34.84%					
AutoStat + ARIMA (fwd)	116,551	158,621	31.25%	37,115,588,103	14,973,85
9,599 0.403 109,667 21.94%					

=====

W12: 2015-07-31 → 2015-12-31

=====

Pre-gap training span: 1989-01-31 → 2015-06-30 (318 months)
 Post-gap training span: 2016-01-31 → 2019-12-31 (48 months)
 Downstream geometry: train ends 2018-12-31 | test 2019-01-01 → 2019-12-31 (2
 4m pre + 36m post + 12m test)
 Ground-truth variance: 28,517,344,536
 ARIMA fwd order: (1, 1, 2) x (2, 0, 0, 12)
 ARIMA bwd order: (0, 1, 0) x (0, 0, 2, 12)
 AutoStat pipeline: Detrending → Deseasonalizing | ARIMA (0, 1, 2) x (0, 0,
 2, 12)

Oracle baseline (real gap) DS-RMSE= 164,915 DS-MAPE= 34.10%

Method	MAE	RMSE	MAPE	orig_var	pred_var
var_ratio DS-RMSE DS-MAPE					
Linear	231,545	264,568	60.03%	28,517,344,536	31,730,040,797
1.113 155,850 32.11%					
STL + PCHIP	257,957	281,640	57.37%	28,517,344,536	90,017,993,006
3.157 166,429 34.39%					
Spline	422,122	475,704	100.72%	28,517,344,536	89,656,806,154
3.144 220,705 45.49%					
Auto ARIMA (fwd)	75,070	85,487	16.73%	28,517,344,536	8,879,639,836
0.311 161,078 33.29%					
Auto ARIMA (bwd)	378,921	410,910	101.98%	28,517,344,536	989,043,216
0.035 126,268 25.90%					
ARIMA blend	158,086	177,341	41.44%	28,517,344,536	23,713,531,972
0.832 155,231 31.99%					

AutoStat + ARIMA (fwd)	92,698	110,573	23.45%	28,517,344,536	3,787,87
0,133	0.133	158,769	32.83%		

Per-Window Visualisations

```
In [8]: PLOT_BUFFER = pd.DateOffset(months=12) # context shown around the gap
N_PER_WINDOW_PLOTS = 3 # only render the most recent windows

METHODS_TO_PLOT = [
    'Linear',
    'STL + PCHIP',
    'Spline',
    'Auto ARIMA (fwd)',
    'Auto ARIMA (bwd)',
    'ARIMA blend',
    'AutoStat + ARIMA (fwd)',
]

for win_name, gap_start, gap_end in WINDOWS[-N_PER_WINDOW_PLOTS:]:
    res = window_results[win_name]
    gap_mask = res['gap_mask']
    plot_start = gap_start - PLOT_BUFFER
    plot_end = gap_end + PLOT_BUFFER

    fig, axes = plt.subplots(2, 4, figsize=(24, 8), sharex=True, sharey=True)
    axes = axes.flatten()

    for ax, method_name in zip(axes, METHODS_TO_PLOT):
        s_imputed = res[method_name]

        # original (ground truth) in grey dashes
        ax.plot(
            monthly_df[plot_start:plot_end].index,
            monthly_df.loc[plot_start:plot_end, 'arrival_count'],
            color='gray',
            linewidth=1.5,
            linestyle='--',
            label='Ground truth',
            zorder=1,
        )

        # imputed series
        color, ls = METHOD_STYLES[method_name]
        ax.plot(
            s_imputed[plot_start:plot_end].index,
            s_imputed[plot_start:plot_end],
            color=color,
            linewidth=2,
            linestyle=ls,
            label=method_name,
            zorder=2,
        )

        # gap region highlight
        ax.axvspan(gap_start, gap_end, alpha=0.12, color='red')
        ax.axvline(gap_start, color='red', linestyle=':', linewidth=1)
        ax.axvline(gap_end, color='red', linestyle=':', linewidth=1)
```

```

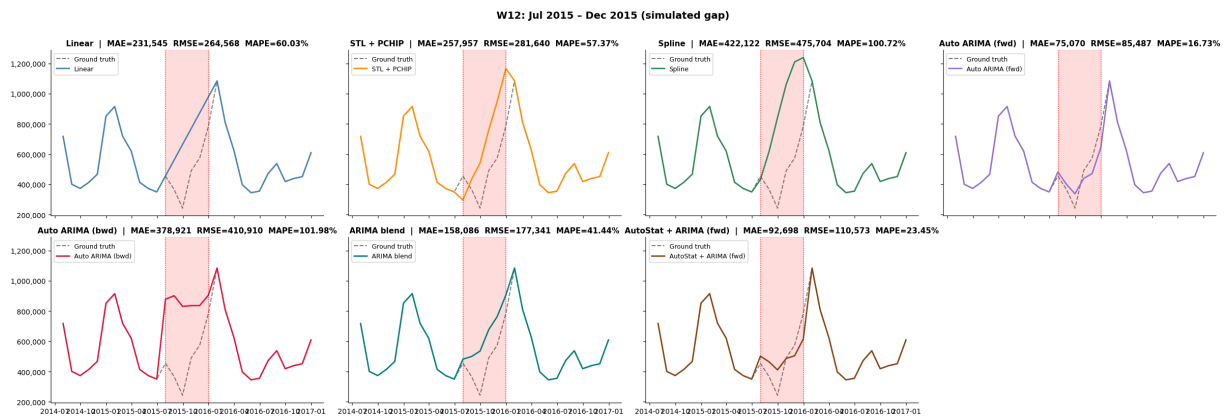
# metrics annotation
win_metrics = results_df[
    (results_df['Window'] == win_name) & (results_df['Method'] == method_name)
].iloc[0]
ax.set_title(
    f'{method_name} | MAE={win_metrics["MAE"]:.0f} | '
    f'RMSE={win_metrics["RMSE"]:.0f} MAPE={win_metrics["MAPE"]:.2f}%',
    fontsize=11,
    fontweight='bold',
)
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _ : f'{int(x):,}'))
ax.legend(loc='upper left', fontsize=9, frameon=True)
sns.despine(ax=ax)

for ax in axes[len(METHODS_TO_PLOT) :]:
    ax.set_visible(False)

fig.suptitle(
    f'{win_name}: {gap_start.strftime("%b %Y")} - {gap_end.strftime("%b %Y")} ('
    fontsize=14,
    fontweight='bold',
    y=1.01,
)
plt.tight_layout()
plt.show()

```





Aggregated Results

```
In [9]: summary = (
    results_df
    .groupby('Method')[
        ['MAE', 'RMSE', 'MAPE', 'VarRatio', 'DownstreamRMSE', 'DownstreamMAPE']
    ]
    .mean()
    .round({
        'MAE': 0,
        'RMSE': 0,
        'MAPE': 2,
        'VarRatio': 3,
        'DownstreamRMSE': 0,
        'DownstreamMAPE': 2,
    })
    .sort_values('DownstreamMAPE')
)

summary.index.name = 'Method'
summary.columns = [
    'Mean MAE',
    'Mean RMSE',
    'Mean MAPE (%)',
    'Mean var_ratio',
    'Mean DS-RMSE',
    'Mean DS-MAPE (%)',
]

# rank by mean Downstream MAPE
summary.insert(0, 'Rank', range(1, len(summary) + 1))

oracle_summary = (
    results_df
    .groupby('Window')[['OracleRMSE', 'OracleMAPE']]
    .first()
    .round({'OracleRMSE': 0, 'OracleMAPE': 2})
)

print(
    f'\nAggregated benchmark (mean across {results_df["Window"].nunique()} windows,
)
```

```

print(
    'var_ratio = var(pred) / var(truth); 1.0 = ideal, <1 over-smooth, >1 over-sp
')
print(summary.to_string())

print(
    '\nOracle baseline per window (auto-ARIMA fit on the real series, same train/te
')
print(oracle_summary.to_string())
summary

```

Aggregated benchmark (mean across 13 windows, sorted by MAPE)

var_ratio = var(pred) / var(truth); 1.0 = ideal, <1 over-smooth, >1 over-spiky

	Rank	Mean MAE	Mean RMSE	Mean MAPE (%)	Mean var_ratio	Me
an DS-RMSE						
Mean DS-MAPE (%)						
Method						
Auto ARIMA (fwd)	1	109768.0	140546.0	26.75	0.526	
99237.0	19.60					
AutoStat + ARIMA (fwd)	2	94292.0	122007.0	23.52	0.505	
103924.0	20.55					
STL + PCHIP	3	265105.0	286821.0	64.03	1.316	
103114.0	20.84					
ARIMA blend	4	159032.0	184203.0	40.30	0.484	
109062.0	21.63					
Linear	5	252464.0	285461.0	64.72	0.471	
114993.0	22.75					
Auto ARIMA (bwd)	6	312016.0	344525.0	77.05	0.610	
127721.0	25.62					
Spline	7	930968.0	1012508.0	222.13	7.850	
215686.0	43.99					

Oracle baseline per window (auto-ARIMA fit on the real series, same train/test geometry):

	OracleRMSE	OracleMAPE
Window		
W0	39783.0	8.75
W1	53576.0	13.11
W10	62349.0	8.17
W11	144430.0	29.85
W12	164915.0	34.10
W2	58258.0	8.97
W3	113316.0	24.39
W4	94891.0	20.76
W5	219691.0	43.92
W6	191332.0	34.35
W7	117749.0	24.05
W8	55954.0	8.13
W9	127319.0	26.13

Out[9]:

	Rank	Mean MAE	Mean RMSE	Mean MAPE (%)	Mean var_ratio	Mean DS-RMSE	Mean DS-MAPE (%)
Method							
Auto ARIMA (fwd)	1	109768.0	140546.0	26.75	0.526	99237.0	19.60
AutoStat + ARIMA (fwd)	2	94292.0	122007.0	23.52	0.505	103924.0	20.55
STL + PCHIP	3	265105.0	286821.0	64.03	1.316	103114.0	20.84
ARIMA blend	4	159032.0	184203.0	40.30	0.484	109062.0	21.63
Linear	5	252464.0	285461.0	64.72	0.471	114993.0	22.75
Auto ARIMA (bwd)	6	312016.0	344525.0	77.05	0.610	127721.0	25.62
Spline	7	930968.0	1012508.0	222.13	7.850	215686.0	43.99

Method Comparison Plot

```
In [10]: N_CHART_WINDOWS = 4 # most recent windows shown alongside the global mean

CHART_METRICS = [
    ('MAE', 'MAE', False, False),
    ('RMSE', 'RMSE', False, False),
    ('MAPE', 'MAPE (%)', True, False),
    ('VarRatio', 'var(pred) / var(truth)', False, True), # ratio panel
    ('DownstreamRMSE', 'Downstream ARIMA RMSE', False, False),
    ('DownstreamMAPE', 'Downstream ARIMA MAPE (%)', True, False),
]

method_order = summary.index.tolist() # already ranked by mean MAPE
window_order = [w[0] for w in WINDOWS] # chronological

fig, axes = plt.subplots(2, 3, figsize=(20, 10))
axes = axes.flatten()

for ax, (col, title, is_pct, is_ratio) in zip(axes, CHART_METRICS):
    pivot = results_df.pivot(index='Window', columns='Method', values=col).reindex(
        window_order
    )[method_order]

    recent = pivot.iloc[-N_CHART_WINDOWS:]
    global_mean = pivot.mean(axis=0, skipna=True).to_frame().T
    global_mean.index = ['Global mean']
    to_plot = pd.concat([recent, global_mean])

    to_plot.plot(kind='bar', ax=ax, width=0.7, edgecolor='white')

    # visually separate the "Global mean" bar from the per-window bars
```

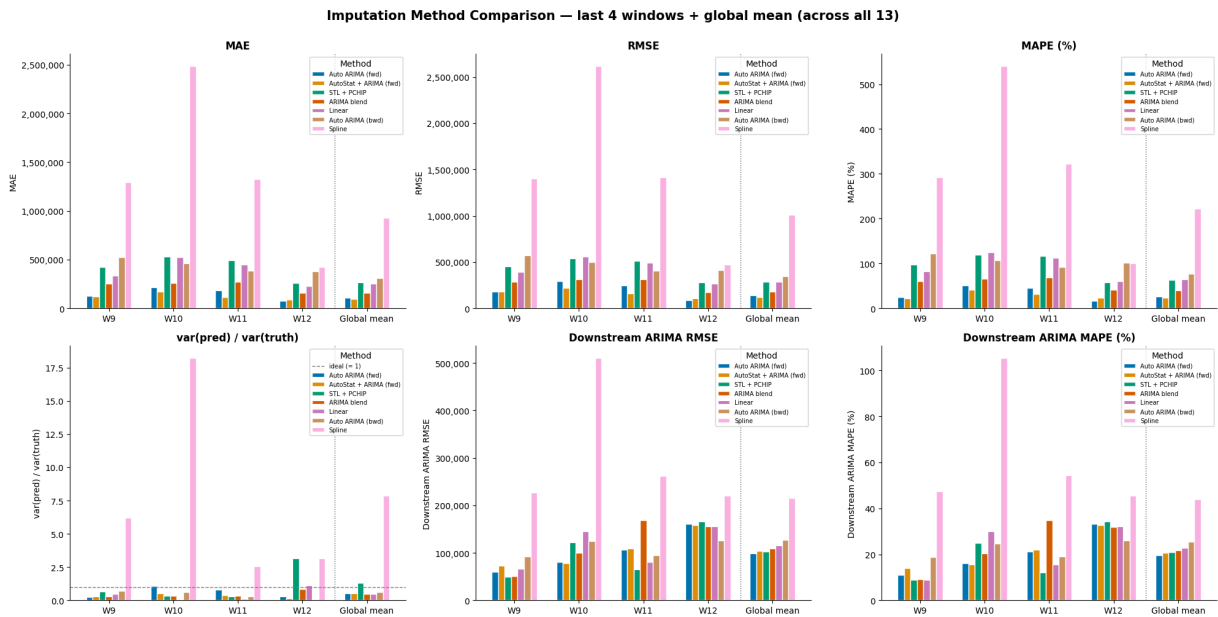
```

ax.axvline(len(recent) - 0.5, color='black', linestyle=':', linewidth=1, alpha=

ax.set_title(title, fontsize=12, fontweight='bold')
ax.set_xlabel('')
ax.set_ylabel(title)
ax.tick_params(axis='x', rotation=0)
ax.legend(title='Method', fontsize=7, loc='upper right')
if not is_pct and not is_ratio:
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
if is_ratio:
    ax.axhline(1.0, color='gray', linestyle='--', linewidth=1, label='ideal (=
handles, labels = ax.get_legend_handles_labels())
    ax.legend(handles, labels, title='Method', fontsize=7, loc='upper right')
sns.despine(ax=ax)

fig.suptitle(
    f'Imputation Method Comparison — last {N_CHART_WINDOWS} windows + global mean (
    fontsize=15,
    fontweight='bold',
    y=1.00,
)
plt.tight_layout()
plt.show()

```



Simple Testing with Pandemic Data

Apply each imputation method (including the AutoML AutoStat + ARIMA) to the **real** COVID gap (2020-01 → 2021-12), then use the resulting fully-imputed series to forecast a **validation year** and compare against its actual values.

The validation year is the **last 12 months of the training data** (the test set of notebook 2 is never used here). It is derived from the data, so it rolls forward when new data is added.

For each method:

1. Mask 2020-01 → 2021-12 and impute it (forward / backward / blend).
2. Fit an auto-ARIMA on the imputed series from 2010-01 up to the month before the validation year.
3. Forecast 12 months → score against the real values of the validation year.

An **oracle baseline** (auto-ARIMA fitted on the same train window but with the *real* COVID values) is included as the upper bound on achievable accuracy.

```
In [11]: N_FORECAST = 12
FORECAST_TRAIN_START = pd.Timestamp('2010-01-01')
# validation year = the last N_FORECAST months of the training data (rolls forward)
FORECAST_TEST_END = monthly_df.index[-1]
FORECAST_TEST_START = monthly_df.index[-N_FORECAST]
FORECAST_TRAIN_END = monthly_df.index[-N_FORECAST - 1]
VALIDATION_LABEL = f'{FORECAST_TEST_START:%b %Y}-{FORECAST_TEST_END:%b %Y}'
print(
    f'Forecast training: {FORECAST_TRAIN_START:%b %Y} → {FORECAST_TRAIN_END:%b %Y}'
    f'validation: {VALIDATION_LABEL}'
)

series = monthly_df['arrival_count'].copy()
covid_mask = (series.index >= COVID_START) & (series.index <= COVID_END)

masked_covid = series.copy()
masked_covid[covid_mask] = np.nan

pre_covid_end = series[series.index < COVID_START].index[-1]
post_covid_start = series[series.index > COVID_END].index[0]

# Impute the real COVID gap with all methods
s_linear_covid = apply_linear(masked_covid)
s_stl_pchip_covid = apply_stl_pchip(masked_covid, covid_mask, pre_covid_end)
s_spline_covid = apply_spline(masked_covid)

s_fwd, fwd_model, forecast_fwd_covid = apply_auto_arima(
    masked_covid, covid_mask, pre_covid_end
)
print(f'Forward ARIMA order: {fwd_model.order} x {fwd_model.seasonal_order}')

s_bwd, bwd_model, forecast_bwd_covid = apply_auto_arima_bwd(
    masked_covid, covid_mask, post_covid_start, FORECAST_TRAIN_END
)
print(f'Backward ARIMA order: {bwd_model.order} x {bwd_model.seasonal_order}')

s_blend = apply_arima_blend(
    masked_covid, covid_mask, forecast_fwd_covid, forecast_bwd_covid
)

s_ast_covid, ast_covid, ast_covid_arima = apply_auto_stationary_arima(
    masked_covid, covid_mask, pre_covid_end
)
print(
    f'AutoStat pipeline: {describe_pipeline(ast_covid)} | '
    f'ARIMA {ast_covid_arima.order} x {ast_covid_arima.seasonal_order}'
)
```

```

)

imputed_variants = {
    'Linear': s_linear_covid,
    'STL + PCHIP': s_stl_pchip_covid,
    'Spline': s_spline_covid,
    'Auto ARIMA (fwd)': s_fwd,
    'Auto ARIMA (bwd)': s_bwd,
    'ARIMA blend': s_blend,
    'AutoStat + ARIMA (fwd)': s_ast_covid,
}

# Real ground truth of the validation year
y_test = series.loc[FORECAST_TEST_START:FORECAST_TEST_END]
assert len(y_test) == N_FORECAST, (
    f'Expected {N_FORECAST} test months, got {len(y_test)}'
)

# Oracle: real COVID values + auto-ARIMA trained up to FORECAST_TRAIN_END
oracle_train = series.loc[FORECAST_TRAIN_START:FORECAST_TRAIN_END]
oracle_forecast = _fit_predict_auto_arima(oracle_train.values, N_FORECAST)
oracle_rmse_val = rmse(y_test.values, oracle_forecast)
oracle_mape_val = mape(y_test.values, oracle_forecast)
print(
    f'\nOracle (real COVID values) RMSE={oracle_rmse_val:>10,.0f}  '
    f'MAPE={oracle_mape_val:>6.2f}%'
)

forecast_records = []
forecasts_val = {'Oracle': pd.Series(oracle_forecast, index=y_test.index)}

print(f'\n {"Method":<20} {"RMSE":>10} {"MAPE":>8}')
for method_name, s_imputed in imputed_variants.items():
    train = s_imputed.loc[FORECAST_TRAIN_START:FORECAST_TRAIN_END]
    forecast = _fit_predict_auto_arima(train.values, N_FORECAST)
    forecast = np.clip(forecast, 0, None)

    r = rmse(y_test.values, forecast)
    m = mape(y_test.values, forecast)
    forecast_records.append({'Method': method_name, 'RMSE': r, 'MAPE': m})
    forecasts_val[method_name] = pd.Series(forecast, index=y_test.index)
    print(f' {"method_name":<20} {"r":>10,.0f} {"m":>7.2f}%' )

forecast_results_df = (
    pd.DataFrame(forecast_records).set_index('Method').round({'RMSE': 0, 'MAPE': 2})
)
forecast_results_df

```

Forecast training: Jan 2010 → Aug 2024 | validation: Sep 2024–Aug 2025
Forward ARIMA order: (2, 1, 2) x (1, 0, 0, 12)
Backward ARIMA order: (0, 1, 0) x (0, 0, 1, 12)
AutoStat pipeline: Detrending → Deseasonalizing | ARIMA (2, 1, 2) x (1, 0, 0, 12)

Oracle (real COVID values) RMSE= 315,692 MAPE= 26.07%

Method	RMSE	MAPE
Linear	338,880	25.15%
STL + PCHIP	327,293	30.61%
Spline	389,791	38.68%
Auto ARIMA (fwd)	352,792	30.75%
Auto ARIMA (bwd)	324,110	23.00%
ARIMA blend	347,360	30.63%
AutoStat + ARIMA (fwd)	346,371	30.74%

Out[11]: **RMSE MAPE**

Method		
Linear	338880.0	25.15
STL + PCHIP	327293.0	30.61
Spline	389791.0	38.68
Auto ARIMA (fwd)	352792.0	30.75
Auto ARIMA (bwd)	324110.0	23.00
ARIMA blend	347360.0	30.63
AutoStat + ARIMA (fwd)	346371.0	30.74

```
In [12]: POST_COVID_TRAIN_START = pd.Timestamp('2022-01-01')

post_covid_train = series.loc[POST_COVID_TRAIN_START:FORECAST_TRAIN_END]
print(
    f'Post-COVID train span: {post_covid_train.index.min().date()} → '
    f'{post_covid_train.index.max().date()} ({len(post_covid_train)} months)'
)

post_covid_forecast = _fit_predict_auto_arma(post_covid_train.values, N_FORECAST)
post_covid_forecast = np.clip(post_covid_forecast, 0, None)

post_covid_rmse = rmse(y_test.values, post_covid_forecast)
post_covid_mape = mape(y_test.values, post_covid_forecast)
print(
    f'\nPost-COVID only (no imputation)   RMSE={post_covid_rmse:>10,.0f}   '
    f'MAPE={post_covid_mape:>6.2f}%'
)

forecasts_val['Post-COVID only'] = pd.Series(post_covid_forecast, index=y_test.index)
forecast_results_df.loc['Post-COVID only'] = [
    round(post_covid_rmse),
    round(post_covid_mape, 2),
]
```

```

]

# ---- Single chart: training (from 2019) + COVID-19 gap imputations + validation -
PLOT_START = pd.Timestamp('2019-01-01')

forecast_styles_ext = {
    'Oracle': ('black', '-'),
    'Linear': METHOD_STYLES['Linear'],
    'STL + PCHIP': METHOD_STYLES['STL + PCHIP'],
    'Spline': METHOD_STYLES['Spline'],
    'Auto ARIMA (fwd)': METHOD_STYLES['Auto ARIMA (fwd)'],
    'Auto ARIMA (bwd)': METHOD_STYLES['Auto ARIMA (bwd)'],
    'ARIMA blend': METHOD_STYLES['ARIMA blend'],
    'AutoStat + ARIMA (fwd)': METHOD_STYLES['AutoStat + ARIMA (fwd)'],
    'Post-COVID only': ('goldenrod', '-'),
}

fig, ax = plt.subplots(figsize=(18, 7))

ax.axvspan(COVID_START, COVID_END, alpha=0.12, color='red', zorder=0)
ax.axvspan(
    FORECAST_TEST_START, FORECAST_TEST_END, alpha=0.08, color='steelblue', zorder=0
)

trans = ax.get_xaxis_transform()
ax.text(
    COVID_START + (COVID_END - COVID_START) / 2,
    0.97,
    'COVID-19 gap\n(imputed)',
    transform=trans,
    ha='center',
    va='top',
    fontsize=9,
    color='firebrick',
    fontweight='bold',
)
ax.text(
    FORECAST_TEST_START + (FORECAST_TEST_END - FORECAST_TEST_START) / 2,
    0.97,
    'validation\nforecast',
    transform=trans,
    ha='center',
    va='top',
    fontsize=9,
    color='steelblue',
    fontweight='bold',
)

# Ground truth
real = series.loc[PLOT_START:FORECAST_TEST_END]
ax.plot(
    real.index,
    real.values,
    color='gray',
    linewidth=1.6,
    linestyle='--',

```

```

        label='Ground truth (real)',
        zorder=2,
    )

    # Imputation curves over the COVID gap (one anchor point on each side for visual co
    for method_name, s_imputed in imputed_variants.items():
        color, ls = METHOD_STYLES[method_name]
        seg = s_imputed.loc[pre_covid_end:post_covid_start]
        ax.plot(
            seg.index,
            seg.values,
            color=color,
            linewidth=2.2,
            linestyle=ls,
            label=method_name,
            zorder=3,
        )

    # validation forecast per method
    for name, fc in forecasts_val.items():
        color, ls = forecast_styles_ext[name]
        label = None if name in imputed_variants else name
        ax.plot(
            fc.index,
            fc.values,
            color=color,
            linewidth=2.2,
            linestyle=ls,
            marker='o',
            markersize=4,
            label=label,
            zorder=3,
        )

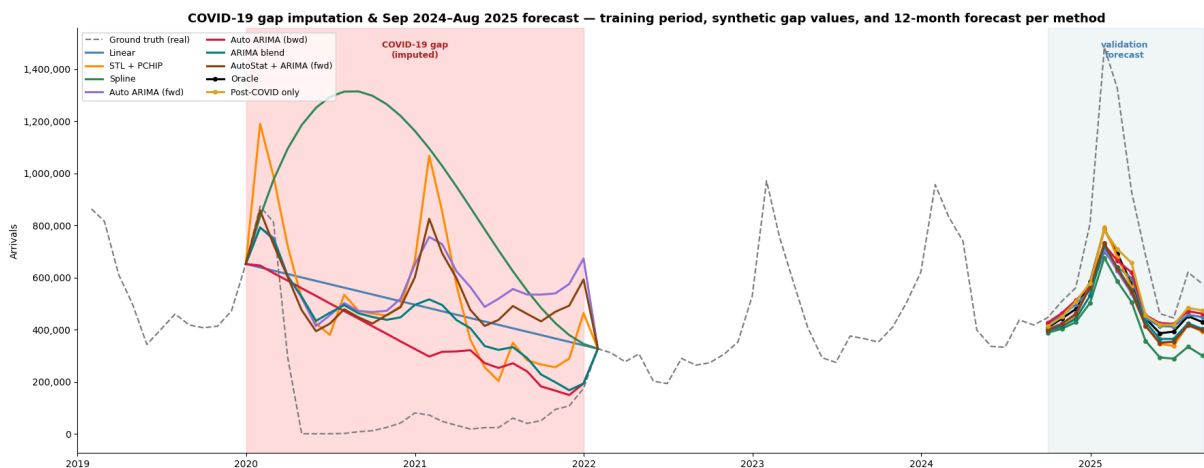
    ax.set_title(
        f'COVID-19 gap imputation & {VALIDATION_LABEL} forecast – training period, '
        f'synthetic gap values, and 12-month forecast per method',
        fontsize=13,
        fontweight='bold',
    )
    ax.set_xlabel('')
    ax.set_ylabel('Arrivals')
    ax.set_xlim(PLOT_START, FORECAST_TEST_END + pd.DateOffset(months=1))
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    ax.legend(loc='upper left', fontsize=9, frameon=True, ncol=2)
    sns.despine(ax=ax)
    plt.tight_layout()
    plt.show()

    forecast_results_df.sort_values('MAPE')

```

Post-COVID train span: 2022-01-31 → 2024-08-31 (32 months)

Post-COVID only (no imputation) RMSE= 301,951 MAPE= 22.42%



Out[12]:

	RMSE	MAPE
Method		
Post-COVID only	301951.0	22.42
Auto ARIMA (bwd)	324110.0	23.00
Linear	338880.0	25.15
STL + PCHIP	327293.0	30.61
ARIMA blend	347360.0	30.63
AutoStat + ARIMA (fwd)	346371.0	30.74
Auto ARIMA (fwd)	352792.0	30.75
Spline	389791.0	38.68

Persist the Imputed Series

The COVID-imputed training series are saved to

`data/processed/train_covid_imputed.parquet` so the modelling notebooks (e.g. notebook 5, ARIMA baseline) can train on a series without the structural break. One column per imputation method; `arrival_count` keeps the real values. Only the training period is saved, so the test set (the last 12 months, split in notebook 2) is never touched.

```
In [13]: imputed_train_df = pd.DataFrame({'arrival_count': series, **imputed_variants})
imputed_train_df.index.name = 'date'
imputed_train_df.to_parquet(
    os.path.join(DATA_FOLDER_PATH, 'train_covid_imputed.parquet')
)

imputed_train_df.loc[COVID_START:COVID_END].round(0)
```

Out[13]:

	arrival_count	Linear	STL + PCHIP	Spline	Auto ARIMA (fwd)	Auto ARIMA (bwd)	ARIMA blend	AutoSta - ARIMA (fwd)
date								
2020-01-31	873295.0	638860.0	1190021.0	834171.0	792362.0	646130.0	792362.0	857701.0
2020-02-29	812726.0	626475.0	988970.0	974854.0	753218.0	617007.0	747296.0	726913.0
2020-03-31	290234.0	613235.0	718523.0	1095584.0	606155.0	587884.0	604566.0	601255.0
2020-04-30	559.0	600423.0	519677.0	1185215.0	522252.0	558761.0	527014.0	475024.0
2020-05-31	517.0	587184.0	424131.0	1251810.0	413328.0	529638.0	433556.0	393939.0
2020-06-30	655.0	574372.0	379528.0	1293068.0	455025.0	500515.0	464914.0	423030.0
2020-07-31	1459.0	561133.0	533852.0	1313826.0	502036.0	471393.0	494042.0	478437.0
2020-08-31	8305.0	547893.0	472461.0	1314469.0	471717.0	442270.0	462755.0	447475.0
2020-09-30	12010.0	535081.0	461438.0	1297959.0	467497.0	413147.0	448593.0	423524.0
2020-10-31	24751.0	521842.0	454465.0	1265281.0	472302.0	384024.0	437758.0	455872.0
2020-11-30	41606.0	509030.0	489805.0	1220540.0	518288.0	354901.0	447250.0	486106.0
2020-12-31	80318.0	495791.0	665571.0	1162837.0	650403.0	325778.0	495148.0	601375.0
2021-01-31	71989.0	482551.0	1067460.0	1095598.0	755858.0	296655.0	516274.0	825555.0
2021-02-28	48096.0	470593.0	854059.0	1028451.0	728059.0	314963.0	494570.0	693237.0
2021-03-31	32439.0	457354.0	571095.0	948982.0	623986.0	316404.0	436762.0	597437.0
2021-04-30	18428.0	444542.0	361362.0	868897.0	563673.0	321103.0	405475.0	476033.0
2021-05-31	23713.0	431303.0	255832.0	784945.0	487294.0	271590.0	337239.0	414203.0
2021-06-30	24070.0	418491.0	202793.0	704537.0	518622.0	253007.0	322298.0	437422.0

	arrival_count	Linear	STL + PCHIP	Spline	Auto ARIMA (fwd)	Auto ARIMA (bwd)	ARIMA blend	AutoSta - ARIMA (fwd)
date								
2021-07-31	60270.0	405251.0	349668.0	624399.0	555801.0	270907.0	332840.0	490621.0
2021-08-31	40410.0	392012.0	282115.0	549380.0	534730.0	239566.0	290899.0	460277.0
2021-09-30	50675.0	379200.0	266355.0	483674.0	534485.0	181938.0	227923.0	431721.0
2021-10-31	93824.0	365961.0	255753.0	424985.0	539095.0	165744.0	198209.0	468646.0
2021-11-30	107123.0	353148.0	288807.0	379096.0	575364.0	148869.0	167412.0	491970.0
2021-12-31	174834.0	339909.0	463479.0	345035.0	672838.0	193440.0	193440.0	592656.0

AutoML Target Transformation for the Validation Forecast

The section above changes *how the COVID gap is filled* and always forecasts with a plain auto-ARIMA ($d=1$, $D=0$) on the raw scale. This section also changes *how the forecaster sees the target*. Every training series is passed through `AutoStationaryTransformer` (Joseph & Tackes, *Modern Time Series Forecasting with Python* 2E) before the auto-ARIMA, and the validation forecast is inverse-transformed back to arrivals.

Two questions:

1. **What does the AutoML pipeline pick for each training series?** Notebook 3 showed that the COVID break masks the trend test and distorts the Guerrero λ . Comparing the pipeline fitted on the *real* series (with the pandemic) against the pipelines fitted on the *imputed* series shows how much each imputation "repairs" the statistical structure the transformer relies on.
2. **Does forecasting in the stationary space beat the raw-scale ARIMA** for the same training series?

Training spans are the same as above: from `2010-01` for the oracle and every imputed series, and from `2022-01` for *Post-COVID only*, both up to the month before the validation year.

```
In [14]: from statsmodels.tsa.stattools import adfuller, kpss # noqa: E402
```

```

training_series = {
    'Oracle': series.loc[FORECAST_TRAIN_START:FORECAST_TRAIN_END],
    **{
        name: s_imputed.loc[FORECAST_TRAIN_START:FORECAST_TRAIN_END]
        for name, s_imputed in imputed_variants.items()
    },
    'Post-COVID only': post_covid_train,
}

diagnostic_rows = []
for name, train in training_series.items():
    ast = fit_auto_stationary(train)
    stationary = ast.transform(train)
    box_cox = [tr for tr in ast._pipeline if isinstance(tr, BoxCoxTransformer)]
    adf_p = adfuller(stationary.to_numpy(), autolag='AIC')[1]
    with warnings.catch_warnings():
        warnings.simplefilter('ignore')
        kpss_p = kpss(stationary.to_numpy(), regression='c', nlags='auto')[1]
    diagnostic_rows.append({
        'Training series': name,
        'n': len(train),
        'Pipeline': describe_pipeline(ast),
        'Trend (MK p)': f'{"✓" if ast._trend_check["trend"] else "X"} '
        f'({ast._trend_check["p_value"]:.4f})',
        'Seasonal (m=12)': '✓' if ast._seasonality_check['seasonal'] else 'X',
        'Heteroscedastic (White p)': (
            f'{"✓" if ast._hetero_check["heteroscedastic"] else "X"} '
            f'({ast._hetero_check["lm_p_value"]:.4f})'
        ),
        'Box-Cox λ': f'{box_cox[0].boxcox_lambda:.3f}' if box_cox else '-',
        'ADF p (output)': round(adf_p, 4),
        'KPSS p (output)': round(kpss_p, 4),
    })

ast_diagnostics = pd.DataFrame(diagnostic_rows).set_index('Training series')
print('AutoStationaryTransformer fitted on each validation-forecast training series')
print(ast_diagnostics.T.to_string())

```

AutoStationaryTransformer fitted on each validation-forecast training series				
Training series	Oracle	Li		
near STL + PCHIP	Spline Auto ARIMA (fwd)	Auto A		
RIMA (bwd)	ARIMA blend AutoStat + ARIMA (fwd)	Post-COVID only		
n		176		
176	176	176	176	
176	176	176	32	
Pipeline	Detrending → Deseasonalizing → AddM → BoxCox	Deseasonalizing	Deseasonalizing	
zoning Deseasonalizing	Deseasonalizing → BoxCox	Deseasonalizing	Detrending → Desea	
sonalizing Deseasonalizing	Deseasonalizing	Deseasonalizing		
Trend (MK p)		✓ (0.0113)	X (0.	
4239)	X (0.3892)	X (0.2151)	X (0.5622)	
✓ (0.0284)	X (0.1574)	X (1.0000)	X (0.2085)	
Seasonal (m=12)			✓	
✓	✓	✓	✓	
✓	✓	✓	✓	
Heteroscedastic (White p)		✓ (0.0029)	X (0.	
2849)	X (0.6174)	✓ (0.0008)	X (0.5957)	
X (0.1451)	X (0.4147)	X (0.7662)	X (0.0900)	
Box-Cox λ			1.567	
—	—	0.082	—	
—	—	—	—	
ADF p (output)			0.2563	0.
1656	0.0	0.0003	0.0	
0.3668	0.0	0.0	0.0076	
KPSS p (output)			0.1	
0.1	0.1	0.1	0.1	
0.1	0.1	0.1	0.0155	

```
In [15]: ast_forecasts_val = {}
comparison_rows = []
raw_scores = {
    'Oracle': (oracle_rmse_val, oracle_mape_val),
    **{
        name: (row['RMSE'], row['MAPE']) for name, row in forecast_results_df.iterrows()
    },
}

for name, train in training_series.items():
    forecast, ast, model = fit_predict_auto_stationary_arma(train, N_FORECAST)
    ast_forecasts_val[name] = pd.Series(forecast, index=y_test.index)
    raw_rmse, raw_mape = raw_scores[name]
    ast_rmse, ast_mape = rmse(y_test.values, forecast), mape(y_test.values, forecast)
    comparison_rows.append({
        'Training series': name,
        'ARIMA order (stationary)': f'{model.order} x {model.seasonal_order}',
        'RMSE raw ARIMA': round(raw_rmse),
        'RMSE AutoStat+ARIMA': round(ast_rmse),
        'MAPE raw ARIMA (%)': round(raw_mape, 2),
        'MAPE AutoStat+ARIMA (%)': round(ast_mape, 2),
        'Δ MAPE (pp)': round(ast_mape - raw_mape, 2),
    })

ast_comparison = (
    pd
    .DataFrame(comparison_rows)
```

```

        .set_index('Training series')
        .sort_values('MAPE AutoStat+ARIMA (%)')
    )
    print(
        f'{VALIDATION_LABEL} forecast – raw-scale auto-ARIMA vs '
        'AutoStationaryTransformer + auto-ARIMA'
    )
    print('Δ MAPE < 0 → the AutoML target transformation improved the forecast\n')
    print(ast_comparison.to_string())

```

Sep 2024–Aug 2025 forecast – raw-scale auto-ARIMA vs AutoStationaryTransformer + auto-ARIMA

Δ MAPE < 0 → the AutoML target transformation improved the forecast

		ARIMA order (stationary)	RMSE raw ARIMA	RMSE AutoStat+ARI
MA	MAPE raw ARIMA (%)	MAPE AutoStat+ARIMA (%)	Δ MAPE (pp)	
Training series				
Post-COVID only		(0, 1, 0) x (0, 0, 0, 12)	301951	2267
25	22.42		-5.24	
Linear		(1, 0, 1) x (2, 0, 1, 12)	338880	2622
65	25.15		-5.46	
ARIMA blend		(1, 0, 0) x (2, 0, 2, 12)	347360	2765
45	30.63		-6.77	
Spline		(1, 0, 0) x (0, 0, 0, 12)	389791	3111
98	38.68		-12.99	
Auto ARIMA (fwd)		(1, 0, 0) x (0, 0, 0, 12)	352792	3111
22	30.75		-3.43	
AutoStat + ARIMA (fwd)		(1, 0, 0) x (0, 0, 0, 12)	346371	3124
99	30.74		-2.53	
STL + PCHIP		(1, 0, 0) x (0, 0, 0, 12)	327293	3026
93	30.61		-1.65	
Auto ARIMA (bwd)		(1, 0, 0) x (2, 0, 2, 12)	324110	3429
92	23.00		11.82	
Oracle		(1, 0, 0) x (0, 0, 0, 12)	315692	3944
08	26.07		17.21	

In [16]: `fig, axes = plt.subplots(1, 2, figsize=(20, 7), gridspec_kw={'width_ratios': [1, 1].`

```

# — Left: MAPE raw vs AutoStat per training series —————
mape_plot = ast_comparison[['MAPE raw ARIMA (%)', 'MAPE AutoStat+ARIMA (%)']]
mape_plot.columns = ['Raw-scale ARIMA', 'AutoStat + ARIMA']
mape_plot.plot(
    kind='barh', ax=axes[0], color=['#9e9e9e', 'saddlebrown'], edgecolor='white'
)
axes[0].invert_yaxis()
axes[0].set_title(
    f'{VALIDATION_LABEL} Forecast MAPE by Training Series', fontweight='bold'
)
axes[0].set_xlabel('MAPE (%)')
axes[0].set_ylabel('')
axes[0].legend(title='Forecaster', frameon=True)
sns.despine(ax=axes[0])

# — right: validation AutoStat forecasts vs ground truth —————
context = series.loc[pd.Timestamp('2022-01-01') : FORECAST_TEST_END]
axes[1].plot(

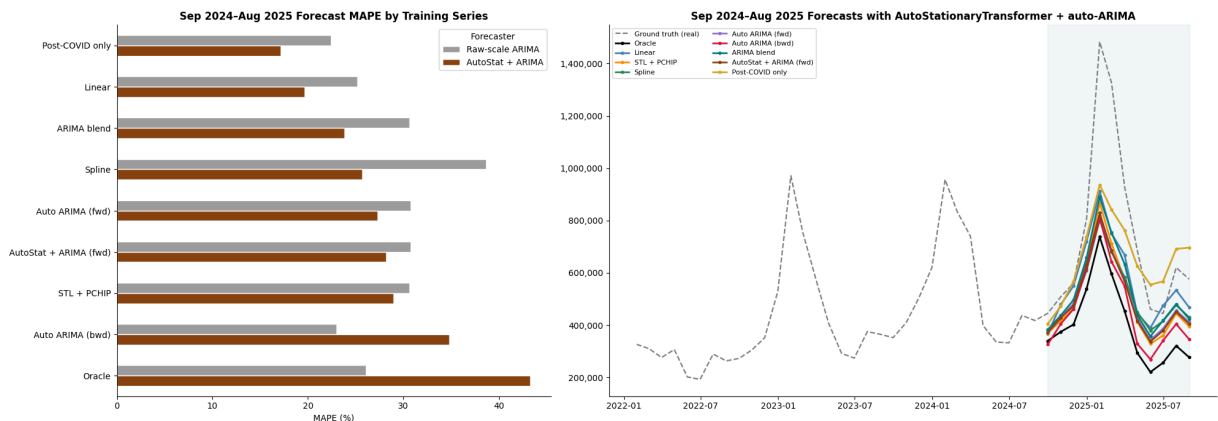
```

```

context.index,
context.values,
color='gray',
linestyle='--',
linewidth=1.6,
label='Ground truth (real)',
)
for name, fc in ast_forecasts_val.items():
    color, ls = forecast_styles_ext[name]
    axes[1].plot(
        fc.index,
        fc.values,
        color=color,
        linestyle=ls,
        linewidth=2,
        marker='o',
        markersize=3,
        label=name,
    )
axes[1].axvspan(FORECAST_TEST_START, FORECAST_TEST_END, alpha=0.08, color='steelblue')
axes[1].set_title(
    f'{VALIDATION_LABEL} Forecasts with AutoStationaryTransformer + auto-ARIMA',
    fontweight='bold',
)
axes[1].yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
axes[1].legend(fontsize=8, ncol=2, frameon=True, loc='upper left')
sns.despine(ax=axes[1])

plt.tight_layout()
plt.show()

```



AutoML Approach — Conclusions

1. As an imputation method (13 simulated pre-COVID gaps)

- *AutoStat + ARIMA (fwd)* gives the **best gap reconstruction** of all seven methods: mean MAPE 23.5% vs 26.8% for the plain forward ARIMA, with the lowest mean MAE and RMSE. It wins in 10 of 13 windows, especially the longer (18–36 month) gaps, e.g. W6: 10.3% vs 19.9% and W11: 31.3% vs 45.4%.

- It loses on the shortest or oldest gaps (W0, W1, W12). There, a straight forward ARIMA over 6–12 months is already accurate. Extrapolating a linear trend fitted on the whole pre-gap history (from 1989) adds bias.
- On the **downstream** metric it is essentially tied with the forward ARIMA (mean DS-MAPE 20.6% vs 19.6%). DS scores differ by less than one point in 8 of 13 windows, because the downstream model trains mostly on real data around the gap. The forward ARIMA is clearly better in W1, W2, W7 and W9 (2–5 pp).
- The pipeline the transformer picks is consistent with notebook 3. It always detrends and deseasonalizes. It adds Box-Cox when the pre-gap history is long and heteroscedastic (W1–W9), and skips it on the short, recent 2014–2015 windows. auto-ARIMA then still chose `d = 1` in 8 of 13 windows (W5–W12), which confirms the **stochastic trend** finding: a deterministic linear detrend alone is not enough.

2. As a target transformation for the 2024 forecast

- **Every imputed series forecasts 2024 better with the AutoML target transformation** than with the raw-scale ARIMA. MAPE improves by 1.9 to 6.5 pp; *Linear* goes from 17.7% to 11.3% and *Spline* from 20.8% to 14.8%.
- The **oracle** (real COVID values) gets much **worse** (18.2% → 35.0%). With the pandemic collapse in the training data, the Mann-Kendall test finds a spurious trend, the Guerrero λ goes to 1.46, and the output is still non-stationary (ADF $p \approx 0.25$). This is the same failure notebook 3 showed on the full 2000–present series: the automatic tests are only as good as the regime they are fitted on.
- On the imputed series, the transformer finds no significant trend (MK $p > 0.05$) and removes the seasonality, and the output is stationary (ADF $p < 0.01$). Imputing the gap is what makes the AutoML pipeline usable.
- *Post-COVID only* remains a strong baseline (11.3% with AutoStat, 11.8% raw), with only 24 months of training data. With AutoStat, *Linear* imputation (11.27%) reaches the same accuracy while keeping 14 years of history for models that need longer inputs (XGBoost lags, PatchTST context windows).

Recommendation

- Do **not** fit an automatic stationarity pipeline on data that contains the raw pandemic period. Impute the COVID gap first, then fit the transformer on the imputed training window only.
- For downstream models, use AutoStat + auto-ARIMA as a candidate both for filling the gap and as the target transformer. Keep the forward ARIMA and *Post-COVID only* as baselines, since the differences on the 12-month 2024 test are a few percentage points on a single year.